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Sustainable valorization of olive stone by-products: Opportunities and challenges

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ABSTRACT

Background: The olive stone, a primary by-product of olive oil extraction, is mainly composed of a lignified shell and inner seed. It represents a substantial portion of the olive industry's biomass waste, contributing over 40 Mt annually. While typically regarded as waste, olive stones contain a variety of nutrients and bioactive compounds like lipids, proteins, phenolic compounds, and minerals found in the seed, as well as fibers in the shell. These elements hold significant value across multiple sectors, including food, energy, and agriculture. These phenolic compounds and nutrients provide notable antioxidant, anti-inflammatory, chemopreventive, and antimicrobial effects, supporting health and disease prevention.

Scope and approach: This review explores the sustainable utilization of olive stone by-products, highlighting their potential to contribute to human health and environmental sustainability. It discusses the practical applications of olive stones in various domains, from functional ingredients in food products and pharmaceuticals to renewable energy sources and soil-enhancing agricultural inputs.

Key findings and conclusions: Olive stones, particularly olive seeds, are rich in dietary fiber (47.6 %), lipids (30.4 %), proteins (13.5 %), and phenolic compounds (8.10 %), especially nüzhenide, nüzhenide 11-methyl

Abbreviations: ABTS, 2,2′-azino-bis3-ethylbenzothiazoline-6-sulfonic acid; ACE, Angiotensin-Converting Enzyme; AChE, Acetylcholinesterase; ADH, Alcohol Dehydrogenase; Akt, Protein Kinase B; ALP, Alkaline Phosphatase; ALT, Alanine Transaminase; AP-1, Activator Protein 1; AST, Aspartate Transaminase; Bax, Bcl-2-associated X protein; BChE, Butyrylcholinesterase; Bcl-2, B-cell lymphoma 2; CAT, Catalase; COX-2, Cyclooxygenase-2; CRP, C-reactive protein; CSE1L, Chromosome Segregation 1 Like; DPP-IV, Dipeptidyl peptidase-IV; DPPH, 2,2-diphenyl-1-picrylhydrazyl; EGFR, Epidermal Growth Factor Receptor; ENOS, Endothelial Nitric Oxide Synthase; ERK, Extracellular Signal-Regulated Kinase; ET-1, Endothelin-1; FA, Fatty Acid; FAO, Food and Agriculture Organization; FAS, Fatty Acid Synthase; FOXO3a, Forkhead Box O3; GC-MS, Gas Chromatography-Mass Spectrometry; GLP-1, Glucagon-like Peptide-1; GSH, Glutathione; H_2O_2, Hydrogen Peroxide; HbA1c, Glycated Hemoglobin A1c; HCD, High Cholesterol Diet; HDAC, Histone Deacetylase; HDL-C, High-Density Lipoprotein-Cholesterol; HFD, High-Fat Diet; HFFD, High-Fat/High-Fructose Diet; HIF-1α, Hypoxia-Inducible Factor 1-alpha; HMG-CoA, 3-Hydroxy-3-Methylglutaryl-CoA; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; HS-SPME-GC-MS, Headspace Solid-Phase Microextraction Coupled with Gas Chromatography-Mass Spectrometry; IFN-γ, Interferon-Gamma; IKKα/β, IκB Kinase Alpha/Beta; IL-6, Interleukin-6; INOS, Inducible Nitric Oxide Synthase; IκBα, Inhibitor kappa B-alpha; JNK, c-Jun N-terminal Kinase; LDH, Lactate Dehydrogenase; LDL-C, Low-Density Lipoprotein-Cholesterol; LOX, Lipoxygenase; LPS, Lipopolysaccharide; MAPK, Mitogen-Activated Protein Kinase; MCP-1, Monocyte Chemoattractant Protein-1; MDA, Malondialdehyde; MMP, Matrix Metalloproteinase; MRSA, Methicillin-Resistant Staphylococcus aureus; MTOR, Mammalian Target of Rapamycin; MUFA, Monounsaturated Fatty Acid; NADPH, Nicotinamide Adenine Dinucleotide Phosphate; NAFLD, Non-Alcoholic Fatty Liver Disease; NF-κB, Nuclear Factor Kappa B; NK cells, Natural Killer Cells; NLRP3, NOD-like Receptor Family Pyrin Domain Containing 3; NO, Nitric Oxide; PDGF, Platelet-Derived Growth Factor; PI3K, Phosphatidylinositol 3-Kinase; PPARα, Peroxisome Proliferator-Activated Receptor Alpha; PUFA, Polyunsaturated Fatty Acid; ROS, Reactive Oxygen Species; SCFA, Short-Chain Fatty Acid; SFA, Saturated Fatty Acid; SOD, Superoxide Dismutase; STAT, Signal Transducer and Activator of Transcription; TAG, Triacylglycerol; TBARS, Thiobarbituric Acid Reactive Substances; TC, Total Cholesterol; TEAC, Trolox Equivalent Antioxidant Capacity; TGF-β, Transforming Growth Factor-β; TNF-α, Tumor Necrosis Factor-α; TXA2, Thromboxane A2; TXB2, Thromboxane B2; VCAM-1, Vascular Cell Adhesion Molecule 1; VEGF, Vascular Endothelial Growth Factor; VLDL-C, Very Low-Density Lipoprotein-Cholesterol; VOC, Volatile Organic Compound.

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oleoside and methoxynüzhenide, and demonstrate a range of health-promoting properties. Additionally, they are shown to benefit metabolic health by combating disorders such as diabetes, hyperlipidemia, obesity, and cardiovascular and neurodegenerative diseases while also protecting organ functions like those of the liver and kidneys. The review underscores the promise of olive stone by-products as a sustainable, health-benefiting resource in circular economy practices within the olive oil industry.

1. Introduction

The olive stone, a by-product of olive and olive oil extraction, is obtained through various processing methods. In the pitted table olive processing industry, whole olive stones, comprising both the endocarp and seed, are collected. In contrast, the olive oil extraction process separates stones during the initial phase, which is called crushing. This process includes stages such as milling the olives, malaxation and using a horizontal centrifuge for oil separation. The two-phase extraction method is the most widely used in olive mills, yielding a semi-solid by-product known as pomace (or *alpeorujo*), which contains non-oily components like depleted pulp, peel, stone fragments, and some residual oil (Bai et al., 2023; García Martín et al., 2020; Padilla-Rascón et al., 2020).

The separation of crushed stones from two-phase olive pomace is now common practice, involving centrifugation to separate crushed stones and then solvent extraction to recover residual oil from the olive pulp before extracting olive pomace oil. Both the stones and extracted pomace are dried and can be used separately (Miranda et al., 2019). To enhance the value of olive pomace, various technologies have been developed with an emphasis on waste management, environmental protection, and the economic benefits for mills. The extraction process includes pitting and collecting stones from virgin and de-oiled spent pomace, which together constitute approximately 20 % of the olive's weight (Bai et al., 2023; García Martín et al., 2020; Padilla-Rascón et al., 2020). Mediterranean countries produce more than 3 Mt of olive oil each year, generating over 4 Mt of olive stones as a by-product. Overall, olive oil extraction produces significant waste biomass—around 40 Mt annually—including more than 20 Mt of dry pomace and 10–30 million Mm^3 of solid waste from olive mills (Bai et al., 2023; Enaime et al., 2024; Osei et al., 2022).

While traditionally viewed as undesirable waste, olive stones offer notable advantages due to their low cost and abundance, making them attractive for various applications (Padilla-Rascón et al., 2020). They hold potential across several industrial sectors, including food, cosmetics, pharmaceuticals, energy, chemical industry, and agricultural valorization (Bai et al., 2023; Bolek, 2020; Espadas-Aldana et al., 2021; Lustosa Filho et al., 2024; Rodrigues et al., 2015; Rodríguez et al., 2008; Trubetskaya et al., 2023).

Additionally, olive stones are rich in bioactive compounds with considerable health-promoting properties, such as antioxidant, anti-inflammatory, chemopreventive, and antimicrobial effects (Batçıoğlu et al., 2023; Celik et al., 2014; Genedy et al., 2020; Kishikawa et al., 2015; Xie et al., 2021; Younis et al., 2023). Major phenolic compounds, like nüzhenide, along with minor phenolic compounds, such as hydroxytyrosol, tyrosol, oleuropein, ligstroside, and rutin have demonstrated potential in supporting cardiovascular and neuroprotective health, as well as in managing metabolic disorders like diabetes, obesity, hepatic steatosis, and kidney diseases (Akhtar et al., 2022; Ben Saad et al., 2021; Cortés-Castell et al., 2016; Genedy et al., 2020; Veciana-Galindo et al., 2015).

This review examines the sustainable use of olive stone by-products, focusing on their nutritional value, health benefits, potential for energy production, and applications in agriculture. By evaluating these diverse uses, this review underscores the promise of olive stones as a sustainable and health-enhancing resource within circular economy practices in the olive oil industry.

2. Composition, production and quantity of olive stone by-product

2.1. Definition and composition

The olive stone, also known as the olive kernel or pit, represents approximately 20.2–38.2 % of the total weight of olive and is composed of two main parts: the endocarp (outer shell) and the endosperm (seed) (Xie et al., 2021). The endocarp, which constitutes about 9–27 % of the olive fruit, serves as a protective layer surrounding the seed and is particularly rich in lignocellulosic compounds—primarily lignin, cellulose, and hemicellulose (García Martín et al., 2020; Younis et al., 2023). This structural makeup gives the endocarp a rigid quality, making it ideal for applications in industries that require durable biomaterials while primarily exploited for pellet production in bioenergy applications (Mateo et al., 2021; Nakamura et al., 2021; Valvez et al., 2021). The endosperm, comprising 2–3 % of the olive fruit, is where most of the nutritional value is concentrated. It contains essential nutrients, including lipids, proteins, carbohydrates, and minerals such as potassium (K^+), calcium (Ca^{2+}), and magnesium (Mg^{2+}), which support energy storage for germination and growth (García Martín et al., 2020; Younis et al., 2023).

Olive stones are rich in phenolic compounds, notably nüzhenide, and other minor compounds renowned for their antioxidant properties and health benefits. This enhances the olive stone's potential for functional applications in various industries (Younis et al., 2023). The biomass of olive stones has a favorable carbon, oxygen, and hydrogen composition, with low nitrogen and sulfur levels. This makes them beneficial for combustion and minimizing harmful emissions, thereby reducing environmental impact (Bai et al., 2023). In the food industry, olive stones can be used to extract sugars and vanillin through specific processing methods, making them a versatile resource in food and flavor production (Bolek, 2020; Rodríguez et al., 2008). Indeed, the valorization of olive stone by-products for sugar production involves a sequential acid and steam explosion pretreatment that fractionates the lignocellulosic components of the olive stones, resulting in a sugar yield of 83 % of their total sugar content. An acid pretreatment using sulfuric acid at 128°C is initially used to release xylose. Subsequently, the olive stones are exposed to steam explosion for 5 min at 195°C, followed by enzymatic hydrolysis to break down the complex sugars into simpler monosaccharides (Padilla-Rascón et al., 2020). Beyond food, these monosaccharides can then be fermented to produce biofuels, such as ethanol and biogas, as well as biomaterials for food packaging and pharmaceuticals, such as biopolymers and cellulose nanofiber, thus contributing to the biorefinery system (Nakamura et al., 2021). Moreover, olive stones have diverse other applications, including as soil amendments in agriculture (Lustosa Filho et al., 2024), components in animal feed (Arapoglou et al., 2024), and raw materials for cosmetics, nutraceuticals, pharmaceuticals (Rodrigues et al., 2015), activated carbon production (Corral-Bobadilla et al., 2021), gas generation (Trubetskaya et al., 2023), and biodegradable composites and eco-friendly materials (Espadas-Aldana et al., 2021).

2.2. Production and quantity

Olive stone production is predominantly associated with the olive oil industry and is heavily concentrated in the Mediterranean region, where approximately 90 % of the world's olives are cultivated due to favorable

climatic conditions (Fraga et al., 2021). In 2021, global olive tree cultivation reached around 11.6 Mt, yielding an estimated 20 Mt of olives annually (García Martín et al., 2020). Olive stones are a significant by-product of both the olive oil extraction industry, which accounts for 90 % of production, and the table olive sector, which contributes the remaining 10 % (Fraga et al., 2021). The olive oil industry alone generates over 4 Mt of olive stones annually (Bai et al., 2023; Osei et al., 2022), while the table olive industry produces an additional 30,000 metric tons (García Martín et al., 2020).

According to recent FAO statistics for 2022, Spain leads in olive oil production, with an average output of 665,709 metric tons per year. Italy follows in second place with an annual production of 331,038 metric tons of olive oil, and Greece ranks third with 313,300 metric tons (FAO, 2024; Padilla-Rascón et al., 2020). Together, these three countries produce approximately 70 % of the world's olive oil and are responsible for generating over 5 Mt of olive by-products annually (Berbel and Posadillo, 2018; Kounani et al., 2022). Each country produces its own variety of olives depending on its unique geo-climatic conditions. The most common olive cultivars in Spain are Picual, Arbequina, and Hojiblanca (Gómez-Gálvez et al., 2024). Two of the most important varieties in Italy are Frantoio and Leccino (Aguilera et al., 2005). Additionally, Koroneiki and Mastoidis are notable cultivars in Greece (Tourvas et al., 2023).

In 2022, Morocco ranked as the seventh-largest olive oil producer globally and the second producer in Africa, following Tunisia, with an annual output of approximately 181,500 metric tons (FAO, 2024). Olive trees are extensively cultivated in regions such as Fès-Meknès (32 %), Marrakech-Safi (19 %), Oriental (15 %), Tangier-Tétouan-Al Hoceima (12 %), and Beni Mellal-Khenifra (10 %), which play a significant role in the country's agricultural sector (Sánchez-Lozano et al., 2023). Additionally, Morocco has a wide diversity of olive cultivars, with approximately 70–80 % represented by Moroccan Picholine and around ~20 % represented by other varieties, such as Haouzia, Menara, and Meslala (Gagour et al., 2022; Issaoui et al., 2024; Sakar et al., 2022). Interestingly, Moroccan olive cultivars exhibit unique phenotypic and genetic diversity. They are particularly adapted to Morocco's arid climate, which may be challenging for European varieties. Indeed, Moroccan cultivars have specific chilling and heat requirements that differ from those of foreign cultivars. Additionally, they show a longer flowering duration, and the timing of fruit development and ripening also varies significantly compared to some foreign olives. The genetic diversity among these cultivars is notable, as they originate from different breeding programs, leading to variations in traits such as flower bud density and fruit set. The unique characteristics of Moroccan cultivars that may not be present in foreign varieties highlight the importance of local adaptation (Joine and Sakar, 2025; Khlil et al., 2016).

As per the FAO's 2022 report, Morocco's olive production exceeded 1.96 Mt, contributing about 4 % to the global olive oil supply (Donner et al., 2022; FAO, 2024). Since the stone of the olive fruit comprises between 20.2 % and 38.2 % of its total weight (Xie et al., 2021), the estimated annual production of olive endocarp in Morocco is approximately 395,920 to 748,720 metric tons.

The expansion of olive plantations globally has bolstered the olive oil industry, with millions of tons of olives processed each year. However, this increase in production also poses environmental challenges, as the disposal of olive stones as waste can contribute to pollution. Fortunately, the industry actively investigates sustainable solutions to recycle this by-product into diverse industrial applications. This shift not only offers a sustainable alternative for managing olive stone waste but also minimizes the environmental footprint of the industry.

3. Physical and chemical properties of olive stone by-product

This section delves into the physical and chemical properties of the olive stone by-product, presenting its origin, morphology, and chemical composition, including its ash content and trace elements. Understanding these properties provides a scientific basis for exploring their diverse applications and benefits, particularly in fields like biomaterial development, agricultural amendments, and bioenergy production.

3.1. Texture and porosity

The lignocellulosic structure of olive shells, composed of lignin, cellulose, and hemicellulose, imparts them with notable rigidity and density, making them suitable for use in applications that require durable biomaterials (Valvez et al., 2021). The irregular, polygonal morphology of olive shells becomes more defined upon grinding, creating fragmented particles with diverse shapes and porosities. Olive shell cells are connected by thin walls, forming a structural network of large vessels with elliptical cross-sections averaging around 43 μm in diameter. Parenchyma cells, measuring 10–50 μm, are organized in a reticular-lattice pattern, contributing to the shell's compact structure. Tensions induced by shrinkage create fissures aligned with the vessel structure, further affecting its porosity.

Studies have identified three primary types of olive shell particles based on porosity: Type 1, which includes broad, flat shapes with low porosity ($\lambda = 2.25$); Type 2, characterized by quasi-flat sections with moderate porosity ($\lambda = 2.54$); and Type 3, featuring crescent or fragmental 2D shapes with high porosity ($\lambda = 4.20$) (Sibaja et al., 2005).

The multi-fractional porosity of olive shells, influenced by both wood and cambium-rich bark, makes them suitable for forming biocomposites with controlled porosity levels, which can be optimized through varying compaction forces (Sibaja et al., 2005). This adaptability in porosity is particularly valuable in thermal applications and contributes to the material's versatility across diverse industrial uses.

3.2. Moisture content

The moisture content of olive stone by-products plays a key role in their handling, storage, and application suitability (Sibaja et al., 2005). Moisture content, determined as the mass of water relative to the dry mass of the material, is measured by exposing the stones to 105°C for 24 hours (Cuevas et al., 2019). Equilibrium moisture content is achieved when the water vapor concentration around the material stabilizes, halting moisture exchange with the surrounding environment. The equilibrium moisture content of olive stones increases with relative humidity but decreases with rising temperatures (Cuevas et al., 2019; Elbir et al., 2012).

Studies have recorded an average moisture content in olive stones of approximately 17.84 % (wet basis) across different *Olea europaea* cultivars from Spain, Morocco, and Tunisia (Mata-Sánchez et al., 2014). Moreover, the equilibrium moisture content of olive stones ranges from 9.1 % to 11.5 % (dry weight basis) when evaluated at 303 K under relative humidity levels between 43.2 % and 73.1 % (Cuevas et al., 2019). Research by Elbir et al. (Elbir et al., 2012) found that Moroccan olive stone by-products reach moisture equilibrium after four hours, with an initial rapid adsorption rate of 0.03283 % per minute, slowing to 0.00478 % per minute as surface sites become saturated (Elbir et al., 2012). This high initial rate and subsequent reduction demonstrate the material's high affinity for moisture, which is essential for applications that rely on moisture regulation.

3.3. Ash content

Ash content is a significant chemical property in assessing the utility of olive stone by-products in applications such as soil amendments, structural materials, and bioenergy. High carbon (50.1–53.7 %) and oxygen (38.8–41.9 %) content enhance its calorific potential, while a lower nitrogen and sulfur content reduces emissions, making it an environmentally friendly biomass option (Mata-Sánchez et al., 2014; Skodras et al., 2006; Vera et al., 2014).

Ash content in olive stones, analyzed according to ASTM D 3174 standards, varies between 0.6 % and 2.6 % across different cultivars of *O. europaea* L. (Blanco López et al., 2002; García et al., 2012; Ghanbari et al., 2012; Mata-Sánchez et al., 2014; Skodras et al., 2006; Skoulou et al., 2008; Vera et al., 2014). The ash composition includes inorganic compounds such as SiO_2 (28.5–31.8 %), K_2O (22.8–32.1 %), CaO (9.3–33 %), Al_2O_3 (3.4–6.4 %), MgO (2.4–5.13 %), and Fe_2O_3 (2.4–7.9 %) (de Moraes Pinheiro et al., 2018; García et al., 2014; Skodras et al., 2006). Factors influencing ash content include the olive variety, soil mineral content, and ripeness stage (de Moraes Pinheiro et al., 2018). Moreover, olive stone by-products exhibit a higher Lower Heating Value (19.15 MJ/kg) and a lower ash content compared to

Table 1
Olive stone chemical composition (*Olea europaea* L.).

Chemical Composition	Concentrations Whole stone	Concentrations Seed	Identification/ quantification methods	References
Total fiber (%)	74.08 %	47.60 %	• HPLC (High-Performance Liquid Chromatographic)	(Maestri et al., 2019; Soares et al., 2024)
Insoluble fiber	-	32.70 %		
Cellulose	20–40 %	-	• HRGC (High-Resolution Gas Chromatography)	
Lignin	25–35 %	-		
Hemicelluloses	18–32 %	-	• LC−MS (Liquid Chromatography-Mass Spectrometry)	
Lipids (%)	-	30.40 %		(Ben Mansour et al., 2015a; Maestri et al., 2019; Ranalli et al., 2002a; Rodríguez et al., 2008)
Oleic acid (C18:1)	-	50–74.4 %	• HPLC-DAD-MS (HPLC with Diode Array Detection and Mass Spectrometry)	
Linoleic acid (C18:2)	-	10–25 %		
Palmitic acid (C16:0)	-	8–16 %		
Stearic acid C18:0	-	2–6 %	• HPLC–ESI–MS/MS (HPLC with Electrospray Ionization and Tandem Mass Spectrometry)	
Eicosenoic acid (C20:1)	-	0.55–0.63 %		
Arachidonic acid (C20:0)	-	0.46–0.56 %		
Linolenic acid (C18:3)	-	0.4–5 %		
Palmitoleic acid (C16:1)	-	0.35–0.43 %		
Behenic acid (C22:0)	-	0.27–0.31 %		
Heptadecenoic acid (C17:1)	-	0.1–0.12 %		
Heptadecanoic (C17:0)	-	0.09–0.11 %		
∑SFA	-	11.83–16.94 %		
∑MUFA	-	62.03–74.08 %		
∑PUFA	-	15.21–25.05 %		
Triterpenoids (mg/kg oil)	-	149.8–921 %		
β-amyrin	-	1530.40–3914.1 %		
Sterols (mg/kg oil)	-	297.00–433.60 %		
β-Sitosterol	-	141.8–162.2 %		
24-Methylenecycloartanol	-	69.1–207.9 %		
Δ^5-Avenasterol	-	0.54–1.25 %		
Campesterol	-	393.79–408.21 %		
Squalene (mg/g oil)	-	8.14–9.96 %		
Tocopherols (mg/kg oil)	-	36.50–41.50 %		
α-tocopherol		8.29–11.51 %		
β-tocopherol				
γ-tocopherol				
δ-tocopherol				
Phenolic content	-	-		(Abdel Rahman et al., 2024; Alu'datt et al., 2011; Ben Mansour et al., 2015b; Cecchi et al., 2023, 2023; Ghanbari et al., 2012; Gouvinhas et al., 2022; Maestri et al., 2019; Servili et al., 1999)
Secroiridoids (mg/g of DW)	3.26 %	9.26–47.46 %		
Nüzhenide	1.32 %	-		
Methoxynüzhenide	0.83 %	3.05–5.62 %		
Nüzhenide 11-methyl oleoside	0.83 %	37.08–91.78 %		
Ligstroside	0.05 %	2.34–2.7 %		
Oleuropein	-	0.1–2.30 %		
Demethyl-oleuropein	-	4.62 %		
Simple phenols (mg/g of DW)	0.05 %	1.05 %		
Hydroxytyrosol	0.10 %	0.49–2.94 %		
Tyrosol	-	-		
Phenolic acids (μg/g of DW)	-	4075.71 %		
Ellagic acid	-	2037.49 %		
Vanillic acid	-	1480.22 %		
Syringic acid	-	1024.78 %		
Chlorogenic acid	-	837.58 %		
Ferulic acid	-	833.34 %		
Caffeic acid	-	1.35–2.58 %		
Flavonoids (mg/g of DW)	-	2.72–3.39 %		
Rutin	-	2.10–2.90 %		
Luteolin−7-*O*-glucoside	1	1.3–14.9 %		
Apigenin 7-*O*-glucoside				
Cyanidin 3-*O*-glucoside				
Cyanidin 3-*O*-rutinoside				
∑Phenolic compounds **(mg/g of DW)**				

MUFA: Monounsaturated fatty acid; PUFA: Polyunsaturated fatty acid; SFA: Saturated fatty acid.

leaves, pulp, and branches (Vera et al., 2014). Compared to other biomass materials, olive stone ash is lower than wood species like Scytalidium (7.5 %) and wheat straw (6.1 %), which enhances its potential as an efficient and sustainable energy source (Sacoto and Grau, 2014; Vera et al., 2014).

Understanding these physical and chemical properties is essential for maximizing the utility of olive stone by-products, enabling industries to leverage their unique characteristics in energy production, biomaterial innovation, and environmentally sustainable applications.

4. Nutritional, biological properties, and health benefits of olive stone

Olive stones, often regarded as by-products, exhibit substantial health benefits and preventive effects against various diseases due to their richness in anti-inflammatory, antioxidant, chemopreventive, and antimicrobial compounds. Indeed, olive stones are a notable source of essential macronutrients like lipids (primarily oleic and linoleic acids) and proteins, as well as micronutrients such as vitamin E and minerals (notably potassium and sodium) (Hannachi et al., 2020; Maestri et al., 2019). Additionally, olive stones contain dietary fibers, including cellulose, hemicellulose, and lignin, which vary in proportion depending on the olive variety and extraction method (Hannachi et al., 2020; Maestri et al., 2019). The phenolic compounds present in olive seeds provide robust anti-inflammatory and antioxidant effects, which can help mitigate inflammation associated with conditions like diabetes and obesity, protect against cellular oxidative stress, and potentially lower risks of cardiovascular, hepatic, and neurodegenerative diseases. Their bioactive components also demonstrate chemopreventive properties by inhibiting cancer cell growth and promoting apoptosis in multiple cancer types (Ben Saad et al., 2021; Genedy et al., 2020; Gouvinhas et al., 2022; Kishikawa et al., 2015; Younis et al., 2023).

Olive stone by-products present a valuable resource for the food, pharmaceutical, and nutraceutical industries, with promising applications in both the health and nutrition sectors.

4.1. Nutritional value

Olive stone by-products are a rich source of macronutrients, micronutrients, and bioactive compounds (Table 1), which we discuss below, including their nutritional benefits and applications.

4.1.1. Dietary fiber content

The dietary fiber composition of olive stones is around 74.08 %, while olive seeds contain approximately 47.6 % on a DW basis (Maestri et al., 2019; Bolek et al., 2020). This significant fiber content has gained considerable interest due to its health and nutritional benefits (Reynolds et al., 2020). The nutritional fiber composition differs according to the olive cultivar. Olive stones contain a mix of cellulose (20–40 %), lignin (25–35 %), and hemicelluloses (18–32 %), along with minimal amounts of pectin. Similarly, olive seeds comprise insoluble fibers, accounting for 32.7 % of their DW (Galanakis, 2011; Maestri et al., 2019; Soares et al., 2024). Cellulose, a glucose polymer linked by β-1,4 glycosidic bonds, is a key component of plant cell walls and is protected by lignin and hemicellulose, making it resistant to breakdown (Fischer et al., 2020; García Martín et al., 2020). Hemicellulose, in contrast, is a more complex polysaccharide with varied sugars (glucose, fructose, xylose, mannose, arabinose, and galactose), which renders it easier to break down than cellulose (García Martín et al., 2020; Scheller and Ulvskov, 2010). Lignin, a polymer comprising monolignols (e.g., *p*-coumaryl, coniferyl, and sinapyl alcohols), imparts chemical and structural durability to lignocellulosic compounds (Dixon and Barros, 2019; García Martín et al., 2020). While olive pulp contains soluble fiber, such as pectin (5–6 g per 100 g of total dietary fiber), seeds contain minimal pectin. The pectic polysaccharides in the pulp have a high degree of esterification (>80 %), contributing to their dietary benefits (Abdel Rahman et al., 2024; Galanakis, 2011).

Olive stone fibers have diverse applications beyond nutrition. In the food industry, powdered olive stone fibers are being explored to enrich products with fiber. Additionally, they are used to create eco-friendly biomaterials and serve as natural exfoliants in cosmetics, enhancing the value of this by-product (Bolek, 2020; García Martín et al., 2020; Rodrigues et al., 2015; Valvez et al., 2021).

4.1.2. Fats profile

Olive stones, particularly their seeds, contain around 30 % oil by DW (Maestri et al., 2019). FA composition varies according to olive variety and the region of cultivation (Kailis, 2017). Typically, oleic acid (C18:1), an ω-9 MUFA, constitutes about 50–68 % of FA, with linoleic acid (C18:2), an ω-6 PUFA, accounting for 10–25 %, and SFAs like palmitic and stearic acids representing 8–16 % and 2–6 %, respectively (Fig. 1) (Hannachi et al., 2020; Maestri et al., 2019; Ranalli et al., 2002a). Indeed, seed and pulp olive oils contain identical species of FA but in different concentrations. Olive seed oil generally has a higher PUFA (linoleic acid) content and a lower MUFA (oleic acid) concentration compared to olive pulp oil (Maestri et al., 2019). Minor FAs, including gondoic, arachidic, behenic, pelargonic, lignoceric, linolenic, palmitoleic, margaric, and heptadecenoic acids, are also present (Hannachi et al., 2020).

Olive stones contain a range of other minor lipid components with potential health benefits and applications in the agri-food industry. These components include various alkanes, triterpenoids, sterols, squalene, phospholipids, and other bioactive molecules. The alkanes present in olive stones, approximately 1.7 % of DW, fall within the C23–C33 range, comprising saturated hydrocarbons that vary by geographic origin, cultivar, and harvest timing. Commonly identified alkanes such as pentacosane (C25), heptacosane (C27), and nonacosane (C29) contribute to the distinctive alkane profile of olive stone oil (Giuffrè, 2021; Rodríguez et al., 2008). This profile serves as a reliable indicator of authenticity, purity, and traceability, critical for quality control in the olive oil sector. Deviations in expected alkane levels or the presence of unusual alkanes can signal potential contamination or adulteration, enhancing quality assurance in the industry (Giuffrè, 2021).

In addition to alkanes, olive stones contain small amounts of triterpenoids, particularly triterpene alcohols and sterols, known for their biological activity. β-amyrin, a primary triterpene alcohol, is found at a concentration of approximately 92.1 mg per 100 g of olive stone oil and is noted for its antioxidant and anti-inflammatory properties, which contribute to potential health benefits (Maestri et al., 2019; Melo et al., 2010; Ranalli et al., 2002b). Furthermore, sterols, another class of bioactive compounds present in olive stones, include β-sitosterol as the most prominent, followed by campesterol, avenasterol, and stigmasterol (Ranalli et al., 2002b). These sterols possess cholesterol-lowering effects, primarily through inhibiting their absorption in the digestive tract (Barkas et al., 2023; Musa-Veloso et al., 2011).

Moreover, olive stone oil contains squalene, a natural antioxidant and a sterol precursor. Although its concentration is lower in olive stones compared to olive fruit and pulp oils, squalene's levels tend to decrease with the olive's maturity index (Ben Mansour et al., 2015a), suggesting that harvesting time may influence its yield. Squalene is of significant interest due to its health-promoting properties, including antioxidant activity that contributes to skin health and protection against oxidative stress. This profile of bioactive components within olive stones highlights their potential value not only as a byproduct of olive oil production but also as a source of health-promoting compounds.

4.1.3. Tocopherols

Tocopherols are fat-soluble compounds with notable antioxidant properties and consist of four main isomers: α-, β-, γ-, and δ-tocopherols. These isomers vary based on the number and position of methyl groups on the phenolic ring, affecting their biological activities (Azzi, 2018).

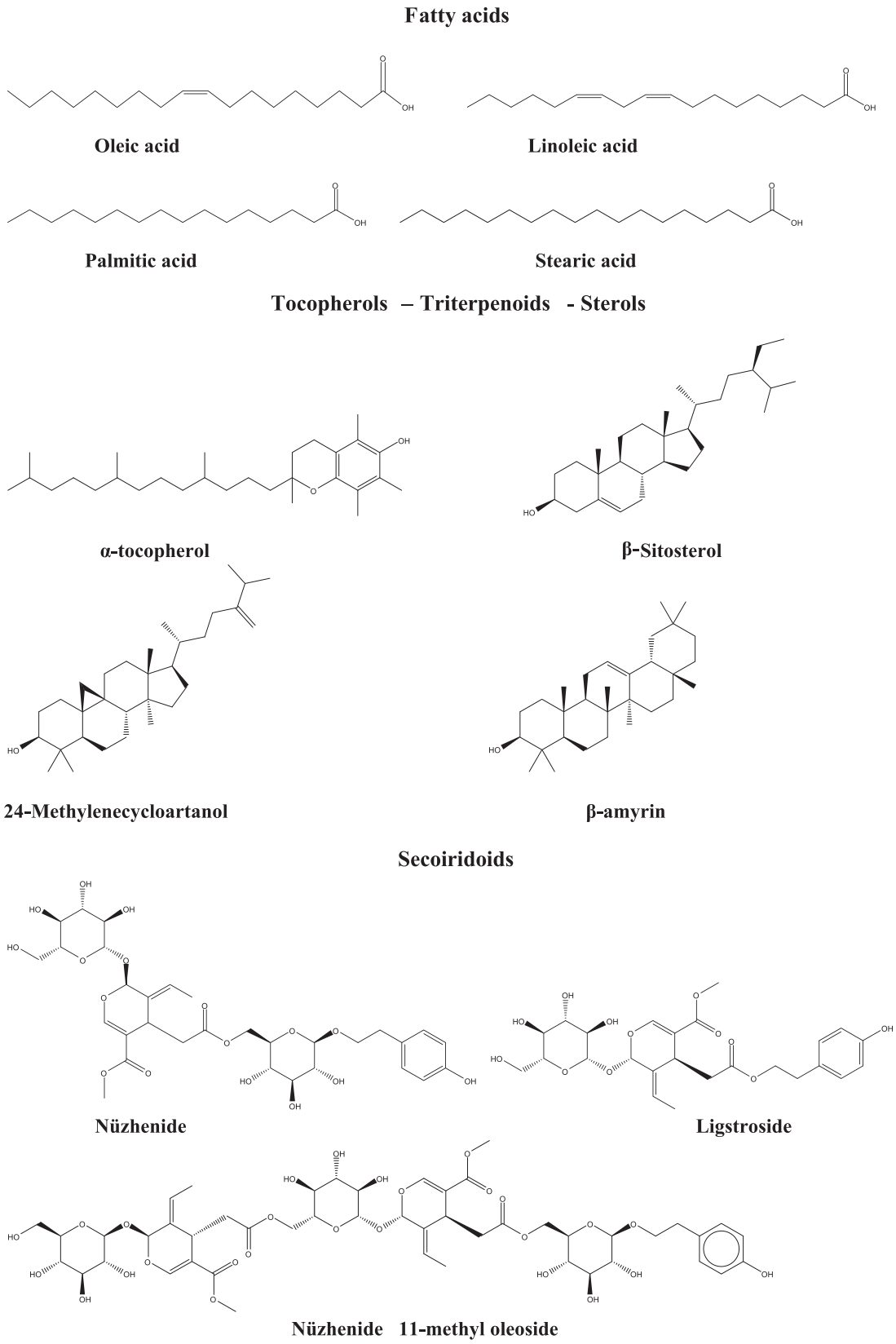


Fig. 1. Chemical structures of major olive stone bioactive compounds: fatty acids, triterpenes, tocotrienols, sterols, and secoiridoids.

Among these, α-tocopherol is the most biologically active and is the only form classified as vitamin E, with recommended daily intakes of 13 mg for men and 11 mg for women (Azzi, 2018; EFSA NDA Panel, 2015). Olive seed oil is especially rich in tocopherols, with α-tocopherol representing approximately 87 % of the total tocopherol content, while smaller quantities of γ-, β-, and δ-tocopherols are also present (Fig. 1) (Maestri et al., 2019).

Studies by Moghaddam et al. (Moghaddam et al., 2012)

demonstrated total tocopherol content in kernel oils from six different *O. europaea* L. cultivars ranging between 15 and 38.8 mg/kg (Moghaddam et al., 2012). By comparison, Maestri et al. (Maestri et al., 2019) found significantly higher levels in Picual cultivars, with a total of 459 mg/kg, of which α-tocopherol accounts for 401 mg/kg, β-tocopherol 9 mg/kg, γ-tocopherol 39 mg/kg, and δ-tocopherol 9.9 mg/kg (Maestri et al., 2019). Tocopherol levels in olive seed oil can vary widely, as seen in 61 cultivars where α-tocopherol content ranges from 122.46 to 520.88 mg/kg (Mousavi et al., 2022). Olive pulp oil also contains substantial α-tocopherol levels, with two Greek cultivars showing concentrations between 382.8 and 562.1 mg/kg, while Italian cultivars average 179.2 mg/kg (Federico Ranalli et al., 2012; Katsoyannos et al., 2015). However, Moghaddam et al. (Moghaddam et al., 2012) found lower tocopherol concentrations in pulp oils, ranging from 11 to 35.8 mg/kg in various cultivars. Additionally, olive pomace oil from various Spanish cultivars contains 357 mg/kg of tocopherols (González-Rámila et al., 2023).

Several factors, including cultivar type, fruit maturity at harvest, agricultural practices, extraction methods, and storage conditions, significantly influence tocopherol content in vegetable oils. Environmental factors like climatic conditions, soil composition, and crop year also play a vital role in tocopherol variation (Rodrigues et al., 2020).

4.1.4. Phenolic content

The phenolic composition of olive stones comprises various compounds, including phenolic alcohols, acids, secoiridoids, and flavonoids, all of which have been thoroughly investigated, especially in olive seeds (Fig. 1).

Olive phenolics exhibit a broad spectrum of health benefits, such as antibacterial, antioxidant, anti-inflammatory, antifungal, antiviral, anti-allergic, and photoprotective effects. Hannachi et al. (Hannachi et al., 2020) analyzed phenolic content in olive stones from various cultivars and oleasters, observing the lowest total polyphenol content (319.08 mg EGA/100 g DW) in olive stones compared to pulp (711.55 mg EGA/100 g DW) and leaves (638.26 mg EGA/100 g DW). They also found lower flavonoid levels (14.75 mg EGA/100 g DW) in stones compared to pulp (23.63 mg EGA/100 g DW) and leaves (19.75 mg EGA/100 g DW) (Hannachi et al., 2020). Significant cultivar-specific differences were noted, with the Chemlali cultivar exhibiting the highest polyphenol and flavonoid levels (Hannachi et al., 2020). The composition of the phenolic content and the methods used for its quantitative analysis in whole olive stones and seeds are listed in Table 1.

Additionally, Maestri et al. (Maestri et al., 2019) identified 24 phenolic compounds in olive seeds, dominated by secoiridoids derivates, along with lower levels of phenolic acids and phenolic alcohols, as well as flavonoids (Maestri et al., 2019). Indeed, unique to olive seeds are compounds such as nüzhenide and its derivatives, especially nüzhenide 11-methyl oleoside and methoxynüzhenide, which are more concentrated in seeds compared to whole olives (Cecchi et al., 2023; Ghanbari et al., 2012; Maestri et al., 2019; Xie et al., 2021). Other phenolics, such as hydroxytyrosol, tyrosol, and oleuropein, can be found in minor concentrations (Ghanbari et al., 2012; Gouvinhas et al., 2022; Maestri et al., 2019). Additionally, Ben Mansour et al. found similar profiles in Chemlali cultivar olive pits, with high concentrations of nüzhenide (325.78 mg/100 g), methoxynüzhenide (132.46 mg/100 g), and nüzhenide 11-methyl oleoside (82.91 mg/100 g), while other minor phenolics included hydroxytyrosol, tyrosol, and oleuropein (Ben Mansour et al., 2015b). Indeed, hydroxytyrosol, tyrosol, and oleuropein are more abundant in whole olives, particularly in the pulp, suggesting that these compounds may migrate during olive processing (Ben Mansour et al., 2015b; Ghanbari et al., 2012).

Flavonoids also represent a substantial portion of olive seed phenolics, with concentrations ranging from 135–258 mg/100 g. Rutin, one of the most prevalent flavonols, can reach concentrations as high as 339 mg/100 g in various Mediterranean cultivars (Gouvinhas et al., 2022). Other flavonoid subclasses include flavones, anthocyanins, and flavanols, featuring compounds like luteolin-7-*O*-glucoside, apigenin-7-*O*-glucoside, and cyanidin-3-*O*-glucoside, among others (Ahmad et al., 2022; Ben Mansour et al., 2015b; Maestri et al., 2019). Lower-abundance compounds such as salidroside, verbascoside, and jaspolyanoside, which are present in small quantities, are also of interest for their health benefits (Ghanbari et al., 2012; Maestri et al., 2019).

Moreover, Maestri et al. (Maestri et al., 2019) highlighted that most phenolic compounds in olive stones exist in glycosylated forms, which contrasts with findings by Alu'datt et al. (Alu'datt et al., 2011), who reported free forms of these compounds. This variation might result from differences in sample preparation or analytical techniques used in the studies.

Recent studies explore the potential applications of olive stone phenolic extracts in functional foods, pharmaceuticals, and cosmetics, owing to their bioactive properties. These compounds may also serve as natural food preservatives, providing an alternative to synthetic antioxidants. Utilizing olive stones for their phenolic compounds not only adds value to agricultural by-products but also supports the sustainability of the olive oil industry by reducing waste (Bolek, 2020; Gullón et al., 2020; Otero et al., 2021).

4.1.5. Protein content

Olive stones, particularly the seeds, are an impressive source of protein, containing between 8.9 % and 18.1 % protein by DW (Galitsopoulou et al., 2022). Mature olive seeds feature a high concentration of storage proteins, with approximately 70 % belonging to the 11S protein family, which supports cellular growth and metabolic functions (Rodríguez et al., 2008). The protein composition is predominantly globulin (50.47 %), followed by albumin (25.72 %), prolamin (17.61 %), and glutelin (6.19 %) (Hannachi et al., 2020). Olive seeds also stand out as an essential source of bioactive peptides, providing several health benefits (Esteve et al., 2015; Prados et al., 2020b). Another study has highlighted the health-promoting potential of these peptides, showing how seed storage proteins can be enzymatically hydrolyzed by enzymes such as Alcalase, Thermolysin, Flavourzyme, Neutrase, and PTN to produce bioactive peptides, particularly those under 3 kDa. These smaller peptides exhibit potent antioxidant and antihypertensive effects, with Alcalase and Thermolysin particularly enhancing these activities (Esteve et al., 2015).

Additionally, these bioactive peptides have demonstrated antiglycemic, antiproliferative, and antimetastatic activities, thus positioning olive seeds as a promising component for functional food applications (Bartolomei et al., 2022; Prados et al., 2020b; Vásquez-Villanueva et al., 2018). The enzymatic hydrolysis process not only enhances their bioactivity but also ensures stability through digestion, reinforcing their potential as a reliable source of health-promoting ingredients (Esteve et al., 2015).

Moreover, olive stones contain a broad spectrum of amino acids, with glutamine (170.45 mg/g protein) as the most abundant non-essential amino acid, followed by asparagine (88.17 mg/g). They also supply all essential amino acids, including valine (162.58 mg/g), leucine (69.66 mg/g), phenylalanine (53.41 mg/g), and lysine (46.63 mg/g). Essential amino acids represent about 46 % of the total amino acid content, and the essential-to-non-essential amino acid ratio (0.85) in olive seed protein surpasses the FAO's recommendation for adults (0.38), which underscores their high nutritional value (Gonçalves et al., 2020; Hannachi et al., 2020; Maestri et al., 2019).

4.1.6. Mineral composition

The mineral profile of olive stones, especially in the seeds, is notably rich. Azbar et al. (Azbar et al., 2004) observed that olive stones have a higher mineral density than the pulp, reaching 4.1 % compared to the pulp's 1–2 %. Potassium is the most abundant macroelement in olive seeds, with approximately 5579 mg/kg, followed by sodium (2758 mg/kg), calcium (2615 mg/kg), magnesium (1878 mg/kg), and

phosphorus (745 mg/kg) (Maestri et al., 2019). Olive seeds are also distinguished by their higher sodium, calcium, and magnesium levels than those found in the whole fruit and pulp, although potassium levels are relatively lower (Fernández-Hernández et al., 2010; Nergiz and Engez, 2000). The phosphorus concentration in olive seeds aligns with the levels found in the entire fruit in some Turkish olive varieties (≈605–1187 mg/kg) (Maestri et al., 2019; Tanilgan et al., 2007). In addition to macroelements, olive seeds contain several microelements vital for human health, including zinc (45.6 mg/kg), manganese (31.5 mg/kg), boron (25.0 mg/kg), copper (23.4 mg/kg), barium (13.4 mg/kg), iron (12.8 mg/kg), chromium (6.3 mg/kg), and nickel (3.9 mg/kg), along with trace amounts of bismuth and cobalt (<0.1 mg/kg) (Maestri et al., 2019). Interestingly, the iron levels in olive seeds are comparable to those in the fruits of various Turkish varieties (9.5–16.5 mg/kg). However, the olive fruit generally has a higher content of chromium (75–219 mg/kg), barium (50–319 mg/kg), and cobalt (8–15 mg/kg), while the seeds show a higher concentration of zinc (3.1–5.2 mg/kg), manganese (1.2–1.9 mg/kg), and copper (2.1–3.4 mg/kg) compared to olive seeds (Fernández-Hernández et al., 2010; Maestri et al., 2019).

4.1.7. Aroma compounds

The aromatic profile of olive seeds, driven by VOCs, contributes to the characteristic fragrance and flavor of olive-derived products, particularly virgin olive oil (Luaces et al., 2003). During extraction processes such as pyrolysis or solvent extraction, hundreds of VOCs can be released, including aldehydes, alcohols, esters, ketones, and terpenes (Luaces et al., 2003; Reboredo-Rodríguez et al., 2013). Olive seeds primarily produce C6 aldehydes, C6 alcohols, and esters, which play a critical role in defining the aroma of the oil (Reboredo-Rodríguez et al., 2013). These compounds are synthesized *via* the LOX pathway from PUFAs, such as linoleic and linolenic acids (Luaces et al., 2003).

A study on the Verdial cultivar from Spain demonstrated that removing olive stones can increase C6 compounds while decreasing C5 compounds, as measured by advanced techniques like GC-MS and HS-SPME-GC-MS (Luaces et al., 2003). The enzymatic activity of ADH in olive seeds specifically targets saturated C6 aldehydes without impacting C5 alcohols. This specificity results in the absence of C5 alcohols in olive seed-derived oils, aligning with the observed absence of ADH activity for these compounds in the Verdial cultivar (*O. europaea* L.) (Luaces et al., 2003).

In contrast, a study by Sakouhi et al. (2020) on Tunisian olive varieties (Chemlali and Ouslati cultivars) found that crushed olive stones contained significantly lower concentrations of volatile compounds compared to the crushed pulp. In fact, over 90 % of the LOX-derived aroma compounds were generated in the pulp, highlighting the pathway's higher activity in the pulp relative to the stones. Furthermore, the Chemlali cultivar exhibited a more intense aroma profile than Ouslati, emphasizing the critical role of the LOX pathway in determining the quality and sensory attributes of olive oil (Sakouhi et al., 2020). Similarly, Reboredo-Rodríguez et al. (2013) reported that in Spanish cultivars (Arbequina, Local, Picual, and Manzanilla de Sevilla), most C6 and C5 volatile compounds associated with extra virgin olive oil aroma originate predominantly from the pulp (80–90 %) rather than the seed (10–20 %). Key volatile compounds indicative of olive ripeness, including trans-2-hexen-1-al, 1-hexanol, trans-3-hexen-1-ol, and cis-3-hexenyl acetate in the pulp and trans-2-hexen-1-al, 1-hexanol, and 3-pentanone in seeds, varied significantly among cultivars, underscoring the influence of cultivar on the aromatic profile and ripening markers in olive oil (Reboredo-Rodríguez et al., 2013).

4.2. Biological properties

Olive stone by-products demonstrate significant health benefits primarily due to their high phenolic content. This section explores the biological activities of these by-products, including their antioxidant, chemopreventive, anti-inflammatory, and antimicrobial properties.

4.2.1. Antioxidant properties

Olive seeds are well-documented for their considerable antioxidant capacities, influenced by factors like seasonal variations, extraction methods, and the presence of bioactive compounds (Table 2). Numerous studies have investigated the antioxidant potential of different seed extracts through various assays, illustrating their capacity to neutralize radicals and combat oxidative stress.

For instance, the ABTS radical scavenging activity was notably higher in aqueous extracts of *O europaea* L. seeds from Japan, reaching 133.4 μg TE/mg, as compared to stems (88.5 μg TE/mg) and fruits (75.5 μg TE/mg), which is closely linked to the phenolic content. Contrastingly, ethanolic seed extracts showed lower ABTS activity compared to leaves and stems, highlighting a weaker antioxidant effect in ethanol-based extracts (Kishikawa et al., 2015). Similarly, Celik et al. (2014) reported that both aqueous and ethanolic extracts exhibited strong antioxidant capacities against ABTS, DPPH, and H_2O_2 radicals at a concentration of 30 μg/mL, which was attributed to their phenolic and flavonoid contents.

Notably, the phenolic content in aqueous seed extracts (46.8 mg GAE/g) was greater than that of other plant parts, such as fruit (43.8 mg GAE/g) and leaves (21–22.6 mg GAE/g) from various *O. europaea* L. varieties (Kishikawa et al., 2015). Seeds from Turkey's Hatay region were also found to possess a higher phenolic composition (317.24 μg GAE) than leaves (280.19 μg GAE) and fruits (270.99 μg GAE). These seeds demonstrated superior Fe^{3+}/Fe^{2+} reducing capacity and a potent DPPH radical scavenging activity (IC_{50} = 5.25 μg/mL) (Batçıoğlu et al., 2023). Other studies have reported ABTS values in the range of 54.03–64.73 μmol TE/g DW for Portuguese seeds and 9–12 μmol TE/g

Table 2
Antioxydant activities of olive seeds of *Olea europaea* L.

Extracts	In vitro models	Results	References
Ethanolic seed extract Ethanolic woody shell extract	Reducing power assay DPPH radical scavenging activity Ferric reducing capacity ABTS radical cation decolorization assay ORAC assay SOSA assay H_2O_2 scavenging activity Metal chelating activity assay (EDTA)	• RPA/IC_{50}= 2.60 ± 0.01 mg/mL • DPPH/IC_{50}= 8.47 ± 0.41 mg/mL • RPA/IC_{50}= 6.11 ± 0.03 mg/mL • DPPH/IC_{50}= 11.57 ± 0.44 mg/mL	(Younis et al., 2023)
Aqueous seed extract		• ABTS= 133.4 ± 18 μg TE/mg • ORAC= 0.17 μmol TE/mg • SOSA= 0.16 ± 0.1 U/μg	(Kishikawa et al., 2015)
Methanolic seed extract Chloroform seed extract Aqueous seed extract N-hexane seed extract		• DPPH/ IC50= 311.9 μg/mL • DPPH/ IC50 = 153.4 μg/mL • DPPH/ IC50 = 118.1 μg/mL • DPPH/ IC50 = 112.5 μg/mL	(Akhtar et al., 2022).
Methanolic seed extract		• DPPH= 3.64–15.93 μmol TE/g • ABTS= 54.03–64.73 μmol TE/g	(Gouvinhas et al., 2022)
Ethanolic seed extract		• DPPH/IC_{50}= 5.25 μg/mL • Fe^{3+}/Fe^{2+}/ IC_{50}= 20.24 μg/mL • H_2O_2/ IC_{50}= 30.50 mg/mL • EDTA/ IC_{50}= 17.37 mg/mL	(Batçıoğlu et al., 2023).

ABTS: 3-ethyl-benzothiazo-line-6-sulfonic acid; ORAC: oxygen radical absorbance capacity; RPA: reducing power assay; DPPH: 2,2-diphenyl-1-picrylhydrazyl; SOSA: superoxide scavenging activity assay.

DW for Italian seeds (Falcinelli et al., 2018; Gouvinhas et al., 2022). Besides, extracts from the Indian olive (*O. europaea* subsp. *Cuspidata*) showed promising antioxidant properties. Akhtar et al. (Akhtar et al., 2022) found that methanol and aqueous seed extracts had IC_{50} values of 311.9 μg/mL and 118.1 μg/mL, respectively, while the methanol extract (500 and 750 mg/kg) significantly reduced MDA levels and increased GSH levels, along with enhancing antioxidant enzymes like CAT and SOD in Wistar rats (Akhtar et al., 2022). In addition, Esteve et al. (Esteve et al., 2015) identified that lower molecular weight peptides in seed hydrolysates, specifically those less than 3 kDa, exhibit strong antioxidant properties. Indeed, the VLDTGLAGA peptide, in particular, demonstrated a high ABTS inhibition rate of 72 % and a TEAC value of 363 ± 28. After gastrointestinal enzyme hydrolysis, the antioxidant capacity of the Alcalase hydrolysate increased to a TEAC value of 382 ± 24 (Esteve et al., 2015).

Moreover, Alu'datt et al. (2011) indicated that optimizing extraction parameters, including time, temperature, solvent choice, sequential extraction, and the use of multi-step extraction methods could significantly enhance the antioxidant capacity of olive seeds.

The antioxidant properties of these seeds are closely related to their total phenolic content, which is assessed *via* mass spectrometry. The antioxidant activity can be attributed to the significant phenolic compound nüzhenide (Cecchi et al., 2023; Servili et al., 1999; Wang et al., 2020). Other phenolic compounds, including ligstroside, tyrosol, luteolin-7-glucoside, oleuropein, and rutin, also contribute to the antioxidant activity of olive stone extracts (Alu'datt et al., 2011; Gouvinhas et al., 2022; Kishikawa et al., 2015; Younis et al., 2023). These compounds are known for their ability to neutralize free radicals and reduce oxidative stress, which plays a critical role in cellular damage and inflammation associated with various chronic diseases (Ahamad et al., 2019; Bulotta et al., 2014; García-Villalba et al., 2015). Oleuropein and hydroxytyrosol, for instance, are effective free radical scavengers, capable of interrupting radical chains and chelating metals, thus reducing oxidative stress. Furthermore, these phenolic compounds can modulate GSH levels and the activities of antioxidant enzymes such as SOD and CAT, leading to a decrease in oxidative stress markers like MDA (Alirezaei et al., 2012; Ivanov et al., 2018; Schaffer et al., 2007).

Oleuropein has been reported to mitigate oxidative stress and decrease TBARS levels by boosting antioxidant enzyme activity in cases of ethanol-induced liver damage in rats (Alirezaei et al., 2012). In bisphenol A-treated rats, oleuropein significantly reduced MDA levels and enhanced CAT and overall antioxidant activities in the liver and kidneys (Mahmoudi et al., 2015). Moreover, oleuropein has been found to be approximately 50 % more effective than vitamin C in a variety of antioxidant assays (Ahamad et al., 2019). Hydroxytyrosol also stimulates phase II detoxification enzymes, which are essential in cellular defenses against oxidative stress (Zhu et al., 2010), and reduces ROS production in low-density lipoproteins, thereby helping to prevent oxidative damage (Martínez et al., 2018). Additionally, nüzhenide and ligstroside activate PPARα, a nuclear receptor that regulates lipid metabolism and oxidative stress-related genes (García-Villalba et al., 2015). Activation of PPARα enhances the expression of key antioxidant enzymes such as CAT and SOD, which help reduce oxidative stress by inhibiting pro-inflammatory pathways like NF-κB, AP-1, STAT, and NLRP3 inflammasome activity (Muzio et al., 2021).

Rutin, another essential phenolic, has been found to mitigate stress by strengthening blood vessel resistance and reducing inflammation. It influences metabolic health by lowering pro-inflammatory markers, including TNF-α, IL-6, COX-2, and IL-1β, while also inhibiting NF-κB and MAPK activation (Muvhulawa et al., 2022). These phenolic compounds not only alleviate inflammation and oxidative stress but also offer synergistic neuroprotection, enhancing cellular resilience to stress.

These findings underscore the potential of olive industry by-products, rich in phenolic compounds, to serve as natural antioxidants. They hold promising applications in the food industry for preserving and enhancing food quality, which aligns with current trends in natural health preservation solutions (Alu'datt et al., 2011).

4.2.2. Chemopreventive activities

Olive seeds display notable antiproliferative and pro-apoptotic activities across a range of cancer cell lines, reinforcing the chemopreventive potential of olive stones (Table 3). These extracts effectively inhibit cancer cell growth, thereby reducing tumor size and proliferation rates. Indeed, Xie et al. (2021) observed that the methanolic extract of seeds from the Frantoio olive cultivar exhibited cytotoxicity against HCT-116 human colorectal cells, with an IC_{50} of 875.5 μg/mL. This indicates that a relatively high concentration of the extract is required to achieve 50 % growth inhibition, suggesting a lower potency relative to other olive extracts, such as branch (IC_{50}=88.25 μg/mL), pomace (IC_{50}=95.85 μg/mL), leaf (IC_{50}=97.06 μg/mL), fruit (IC_{50}=154.3 μg/mL), and olive oil (IC_{50}=170 μg/mL). Indeed, the chemopreventive activity observed in HCT-116 cells can be attributed to phenolic compounds and triterpenoids, particularly ursolic acid, present in the seed extract, which specifically inhibits colorectal cancer cell proliferation.

Furthermore, ethanolic extracts from both seed and woody shells show significant chemopreventive effects on HT-29 (colorectal cancer) and MCF-7 (breast cancer) cell lines. Among these, the seed extract demonstrated the highest activity, with an IC_{50} of 580.96 ± 3.97 μg/mL for HT-29 and an IC_{50} of 881.86 ± 6.22 μg/mL for MCF-7, indicating a stronger effect on HT-29 cells than on MCF-7 cells (Younis et al., 2023). In addition, the LLPSY peptide derived from olive seed hydrolysates exhibited pronounced antiproliferative and antimetastatic effects on prostate (PC-3) and breast (MDA-MB-468) cancer cell lines within 24 hours, with IC_{50} values of 86.1 ± 2.6 μg/mL and 97.6 ± 1.9 μg/mL, respectively. This effect is due to S-phase cell cycle inhibition and a significant decrease in cancer cell migration capacity, with a more marked reduction in breast cancer cell migration (63 %) compared to prostate cancer cells (49 %). However, further *in vivo* experiments are necessary to confirm this antitumor efficacy (Vásquez-Villanueva et al., 2018).

Olive seeds contain high levels of phenolic compounds strongly linked to their chemopreventive activity. Various studies have identified phenolics, primarily nüzhenide, and also hydroxytyrosol, tyrosol, and oleuropein, which exert chemopreventive effects through anti-inflammatory mechanisms, induction of apoptosis, and inhibition of cancer cell proliferation. A study has demonstrated that nüzhenide has chemoprotective activity by inhibiting benzo[*a*]pyrene-induced mutagenicity in bacteria (Niikawa et al., 1993). Indeed, benzo[*a*]pyrene is cleaved by cytochrome P450 into a carcinogenic metabolite known as 7β,8α-dihydroxy-9α,10α-epoxy-7,8,9,10-tetrahydrobenzo[*a*]pyrene.
This compound can form DNA adducts, leading to mutations and malignant changes, and is strongly associated with lung cancer (Bukowska et al., 2022). In addition, hydroxytyrosol, for instance, induces apoptosis in human colon cancer cells (DLD1) by generating ROS and destabilizing redox balance through modulation of the PI3K/Akt/FOXO3a signaling pathway (Fig. 2A) (Sun et al., 2014). This phenolic compound also triggers apoptotic cell death in colorectal adenocarcinoma cells (HT-29) in a p53-dependent manner (Sun et al., 2014). Moreover, hydroxytyrosol can initiate apoptosis in human promyelocytic leukemia and hepatocellular carcinoma cells *in vitro*, leading to G_0/G_1 and G_2/M cell cycle arrests, respectively (Fabiani et al., 2002; Zhao et al., 2014).

Tyrosol, another minor compound, plays a role in preventing oxidative cellular damage that can lead to cancer. It protects L6 muscle cells from H_2O_2-induced cell death by modulating critical signaling pathways, such as ERK, JNK, and p38 MAPK, thereby maintaining cell viability under oxidative stress conditions (Lee et al., 2018). Additionally, tyrosol influences FAS activity, which is implicated in cancer cell growth, through the AMPK/mTOR pathway in human colon cancer cells (Bueno et al., 2019). Hydroxytyrosol also inhibits the expression of COX-2, an enzyme involved in inflammation and cell proliferation, by enhancing ROS levels and modulating prostaglandin and

Table 3
Chemopreventive activities of olive stones of *Olea europaea* L.

Extracts	Cell lines	In vitro models	Results	References
Methanolic seed extract	Human colorectal cancer cell line (HCT−116)	MTS assay for cell viability	IC_{50}= 875.5 μg/mL	(Xie et al., 2021)
Ethanolic seed extract	Human colorectal cancer cell line (HT−29)	SRB assay for cell viability	IC_{50}= 580.96 ± 3.97 μg/mL	(Younis et al., 2023)
	Human breast cancer cell line (MCF−7)	MTT assay for cell viability	IC_{50}= 881.86 ± 6.22 μg/mL	
Ethanolic woody shell extract	Human colorectal cancer cell line (HT−29)	Cell adhesion assay	IC_{50}= 634.81 ± 4.85 μg/mL	(Younis et al., 2023)
	Human breast cancer cell line (MCF−7)	Wound healing assay	IC_{50}= 1720.22 ± 9.06 μg/mL	
		Flow cytometry analysis		
Seed hydrolysates (LLPSY peptide)	Human prostate cancer cell line (PC−3)		IC_{50}= 86.1 ± 2.6 μg/mL	(Vásquez-Villanueva et al., 2018)
	Human breast cancer cell line (MDA-MB−468)		IC_{50}= 97.6 ± 1.9 μg/mL	

MTS: 3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium; MTT: 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; SRB: sulforhodamine B.

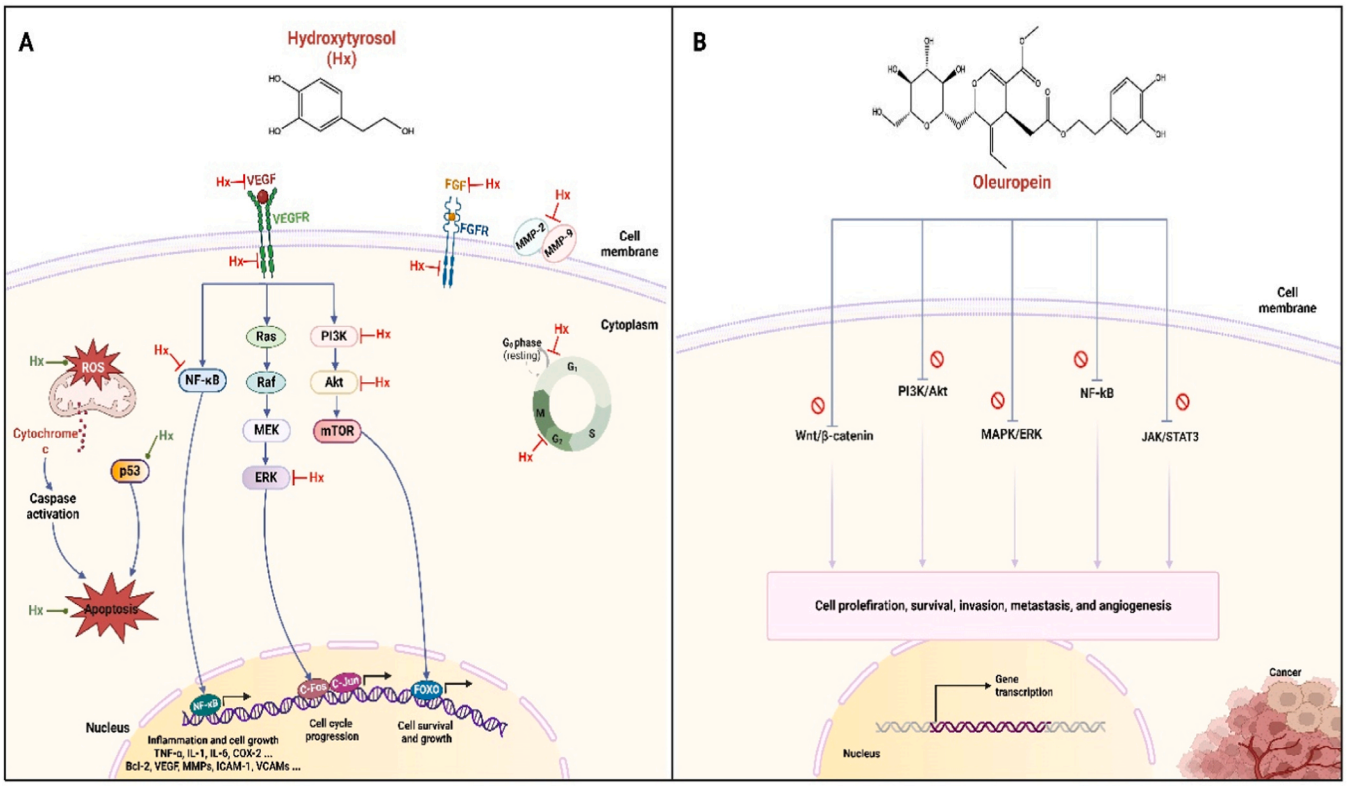


Fig. 2. Chemopreventive molecular pathways of hydroxytyrosol (A) and oleuropein (B). (A) Hydroxytyrosol leads to apoptosis by enhancing the generation of ROS and inactivating the PI3K/Akt/mTOR signaling pathway or through p53 activation. It also can modulate NF-κB and ERK signaling pathways by inhibiting its transcriptional activation and eventually decreasing inflammation and cell proliferation. Thus, hydroxytyrosol can inhibit the gene transcription involved in inflammation and cell proliferation and growth, like COX-2, VEGF, Bcl-2, IL-6, and TNF-α, among others). Hydroxytyrosol can also induce apoptosis by causing G_0/G_1 and G_2/M cell cycle arrest. (B) Oleuropein exhibits significant chemopreventive effects by inhibiting various signaling pathways, including Wnt/β-catenin, P3IK/Akt, MAPK/ERK, NF-κB, and STAT3. AMPK: AMP-activated protein kinase; Akt: protein kinase B; COX2: cyclooxygenase-2; ERK: extracellular signal-regulated kinase; FGF: fibroblast growth factor; FGFR: fibroblast growth factor receptor; FOXO: forkhead box O; ICAM-1: intercellular adhesion molecule 1; IL-6/1: interleukin 6/1; JAK: Janus kinase; MAPK: mitogen-activated protein kinase; MMP: matrix metalloproteinase; mTOR: mammalian target of rapamycin; NF-κB: nuclear factor-kappa B; PI3K: phosphatidylinositol 3-kinase; Ras: rapidly accelerated fibrosarcoma; Ras: rat sarcoma virus; ROS: reactive oxygen species; STAT: signal transducers and activators of transcription; TNF-α: tumor necrosis factor; VCAM: vascular cell adhesion molecule; VEGF: vascular endothelial growth factor; VEGFR: vascular endothelial growth factor receptor.

metalloproteinase activities, as well as various growth factors such as VEGF and PDGF (Desai et al., 2018).

In platelet-rich plasma studies, hydroxytyrosol has been found to reduce the synthesis of TXB2, a marker of TXA_2, which plays an essential contribution in angiogenesis and metastasis (Rufino-Palomares et al., 2022). Additionally, tyrosol demonstrates anti-invasive properties in colon carcinoma cells (HT-115), reducing invasiveness by 30–70 % (Hashim et al., 2008). Hydroxytyrosol and tyrosol further impact the NF-κB signaling pathway by inhibiting its transcriptional activity, which decreases inflammation and cancer cell proliferation. Ye et al. (Ye et al., 2017) demonstrated that tyrosol reduces intestinal inflammation by inhibiting IL-8 secretion in human colon carcinoma cells (WiDr) under oxidative stress by inhibiting NF-κB nuclear translocation. Tyrosol also downregulates lipid synthesis in liver cells, which may reduce the risk of metabolic-related cancers (Sarna et al., 2016). *In vivo* studies support the chemopreventive potential of tyrosol, indicating that it can suppress tumor growth and angiogenesis by inhibiting both Akt and NF-κB signaling pathways (Karković Marković et al., 2019).

Oleuropein also shows significant chemopreventive effects by impairing glycolysis, which reduces cell viability by 30–50 % in human melanoma cells (Ruzzolini et al., 2020). Additionally, it has demonstrated protective effects against colorectal carcinogenesis in mice by

inhibiting key signaling pathways, including Wnt/β-catenin, PI3K/Akt, MAPK, NF-κB, and STAT3. This suppression is accompanied by a reduction in pro-inflammatory cytokines, such as IL-6, IL-17, TNF-α, and IFN-γ, which contribute to its chemopreventive properties, particularly in inflammatory contexts (Fig. 2B) (Giner et al., 2016). Oleuropein also triggers apoptosis in hepatocellular carcinoma cells by modulating cell death-regulating proteins like Bcl-2 and Bax, which are integral components of apoptotic pathways (Yan et al., 2015).

4.2.3. *Anti-inflammatory effects*

Olive seeds exhibit significant anti-inflammatory activities, demonstrating their potential as therapeutic agents against inflammation-related diseases. A comparative study by Kishikawa et al. (Kishikawa et al., 2015) assessed the anti-allergic and anti-inflammatory effects of *O. europaea* L. seeds from various cultivars, including Mission, Manzanillo, Nevadillo Blanco, and Lucca, originating from Japan. Both ethanolic and aqueous seed extracts, tested at a concentration of 400 μg/mL, were evaluated for their ability to stabilize basophils and inhibit inflammatory mediators like histamine and cytokines. These mediators are released from basophils, playing a role in allergic reactions. The experiment involved stimulating RBL-2H3 cells (a basophil analog cell line) with seed extracts for 1 hour, after which granule release was measured. While both extracts demonstrated anti-inflammatory effects, the ethanolic extract showed a stronger inhibition (16.5 %) compared to the aqueous extract (4.4 %), indicating higher efficacy in reducing granule release. In addition, recent findings demonstrated the anti-inflammatory effect of olive pit extracts on the THP1-XBlue-CD14 human leukemia monocytic cell line, specifically targeting the NF-κB-mediated transcriptional pathway, a critical pathway in inflammatory processes. The results indicated that treatment

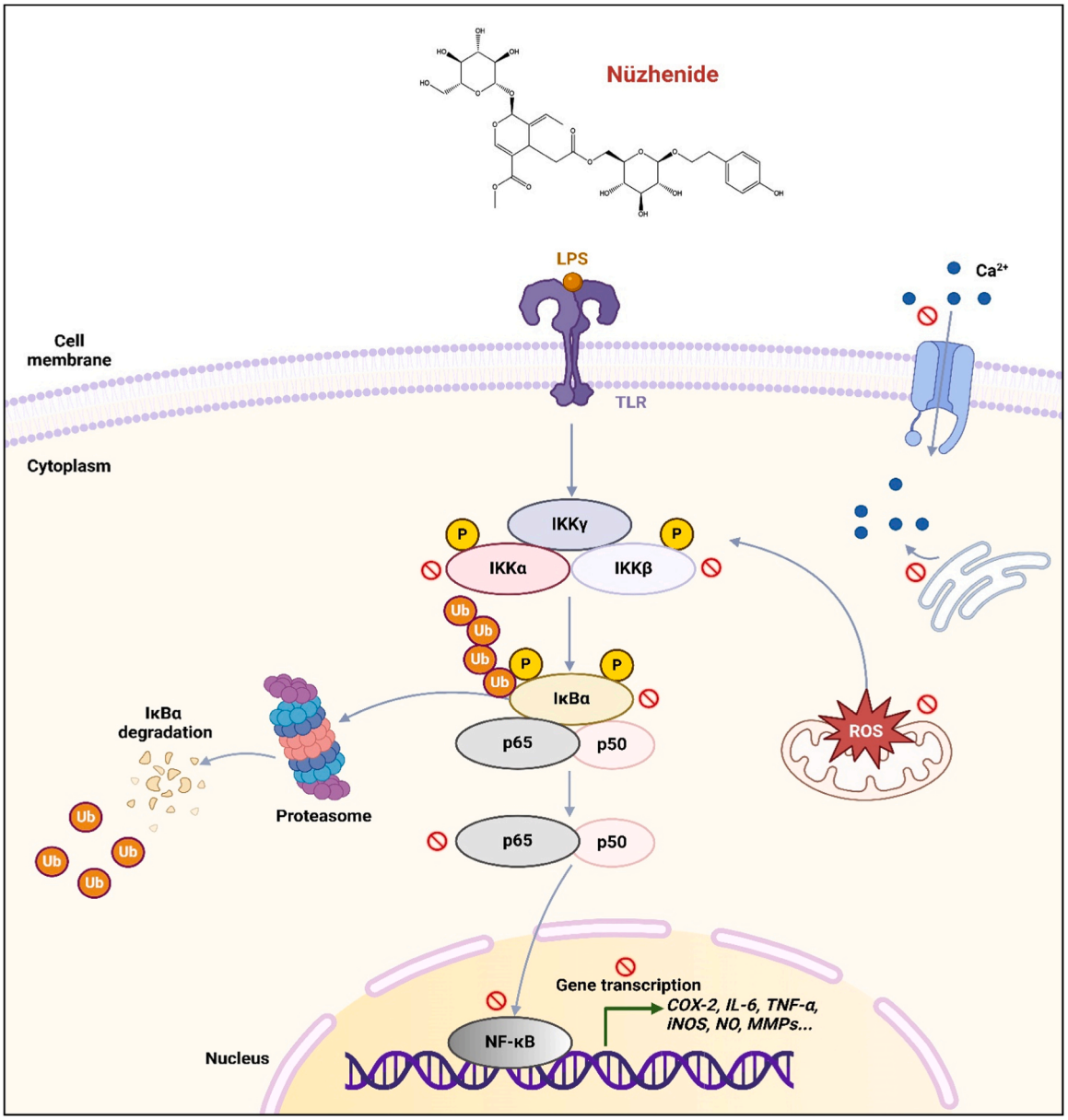


Fig. 3. Anti-inflammatory molecular pathways of nüzhenide. Nüezhenide inhibits the NF-κB pathway in LPS-stimulated RAW264.7 cells by inhibiting the phosphorylation and activation of IKKα/β, IκBα, and p65 proteins, thus suppressing the translocation of p50/p65 complex into the nucleus and the transcription of target genes such as COX-2, TNF-α, IL-6, iNOS, and NO. Nüzhenide also reduces the calcium influx and stabilizes MMPs, leading to a decrease in inflammation. COX-2: Cyclooxygenase-2; IKKα/β/γ: IκB kinase α/β/γ; IL-6: Interleukin 6; LPS: lipopolysaccharide; MMP: Matrix metalloproteinase; NF-κB: Nuclear factor kappa B; NO: NO: Nitric oxide; ROS: Reactive oxygen species; TLR: Toll-like receptor; TNF-α: Tumor necrosis factor-α.

with a 50 mg/L concentration of olive pit extract reduced NF-κB activity by 25.5 %, which correlated with a 34.9 % decrease in inflammatory response compared to untreated cells, thereby illustrating the extract's potential to modulate inflammation effectively (Cortés Castell et al., 2014a).

The phenolic composition of olive seed may play a crucial role in the extracts' anti-inflammatory properties, mainly due to nüezhenide presence. Indeed, this primary phenolic compound exhibited a remarkable anti-inflammatory activity in LPS-stimulated RAW264.7 cells after 24 hours, showing a concentration-dependent effect at levels ranging from 10 to 40 μM. Indeed, Wang et al. (2020) showed that nüzhenide induces an inhibition in the expression of COX-2 and iNOS proteins, resulting in a significant decrease in the levels of inflammatory cytokines such as IL-6, TNF-α, and nitrite. Moreover, nüezhenide can inhibit the transcriptional activity of NF-κB by suppressing the expression of phosphorylated proteins like IKKα/β, IκBα, and p65. Interestingly, nüzhenide also inhibits calcium influx and stabilizes MMPs, thus reducing immune cell activation and reducing inflammation (Fig. 3). Furthermore, other minor phenolic contents like tyrosol, hydroxytyrosol, and oleuropein have demonstrated significant anti-inflammatory activities (Rasouli et al., 2021). Sato et al. (2014) reported that oleuropein aglycone (3,4-DHPEA-EA) and luteolin, phenolic compounds found in the Mission olive cultivar, serve as active anti-allergic agents.

Genedy et al. (2020) investigated the effects of olive seed supplementation in rats fed HFFD, which is known to elevate inflammatory markers. The study used both native and nano forms of olive seed powder, with supplementation levels set at 5 % and 10 %. Results indicated a significant decrease in inflammatory markers, particularly in the group receiving 10 % nano olive seed supplementation. This group exhibited the most notable reduction in serum levels of IL-6 (41.33 ± 5.92 pg/mL compared to 108.72 ± 8.67 pg/mL in untreated rats), CRP (11.80 ± 0.69 ng/mL compared to 216.02 ± 5.51 ng/mL), and MDA (an oxidative stress marker, 18.95 ± 0.69 nmol/L compared to 90.11 ± 6.34 nmol/L). This enhanced anti-inflammatory and antioxidant effect is attributed to the high content of polyphenols and flavonoids in olive seed powder, especially in the nano form, suggesting the potential of nano-processed olive seed powder as a functional food with substantial health benefits for reducing systemic inflammation and oxidative stress.

Overall, the evidence supports the efficacy of olive seed extracts, particularly when concentrated and nano-processed, in reducing inflammation through the inhibition of key inflammatory mediators, modulation of immune cell responses, and suppression of pro-inflammatory pathways such as NF-κB. These findings highlight olive seeds' potential application in functional foods or therapeutic agents aimed at mitigating inflammation-related diseases, warranting further investigation, especially in clinical settings, to confirm their efficacy in human models.

4.2.4. Antimicrobial activities

Olive seed extract demonstrates notable antimicrobial activity, targeting a variety of bacterial and fungal pathogens, which underscores its potential as a natural antimicrobial agent. A recent study focused on olive seed extracts from the Halhalı cultivar in Turkey, assessing their efficacy against several bacterial strains, including *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Enterococcus faecalis*, *Escherichia coli*, *Klebsiella pneumoniae*, MRSA) and *Acinetobacter baumannii*, as well as fungal strains like *Candida albicans* and *Candida glabrata*. Using the agar dilution method, the study indicated that olive seed extracts, at concentrations between 200 and 400 μg/mL, had significant antimicrobial activity, especially against resistant strains like *P. aeruginosa*, *A. baumannii*, and MRSA (Batçıoğlu et al., 2023). Notably, the seed extracts showed stronger antimicrobial effects than fruit extracts but were less effective than leaf extracts, which were active at lower concentrations (100–200 μg/mL) (Batçıoğlu et al., 2023). Complementing these findings, another study highlighted the antimicrobial activity of olive pit extracts at a concentration of 10 mg/mL, which exhibited strong inhibitory effects particularly against *Streptomyces* species (Celik et al., 2014). However, contrasting results were reported by Kishikawa et al. (Kishikawa et al., 2015), who observed no significant antibacterial activity from aqueous and ethanolic olive seed extracts against *S. aureus* and *E. coli* across several Japanese cultivars, suggesting potential variability in antimicrobial efficacy depending on cultivar, extraction method, or concentration.

The antimicrobial efficacy of olive seed extracts appears closely linked to their high phenolic content, which includes compounds like ferulic acid and caffeic acid. These phenols may act synergistically, enhancing the disruption of bacterial cell walls or interfering with critical cellular processes. Phenolic acids are known to target cell wall synthesis and integrity, which is particularly effective against Gram-positive bacteria like *S. aureus*, due to the simpler cell wall structure that allows easier penetration by phenolic compounds (Kishikawa et al., 2015; Younis et al., 2023). Additionally, olive seed extracts contain various lipophilic compounds, which are thought to enhance membrane permeability and disrupt bacterial cell membranes, amplifying antimicrobial activity, especially against Gram-positive bacteria (Kishikawa et al., 2015; Younis et al., 2023).

In summary, the antimicrobial activity of olive seed extracts against both Gram-positive and Gram-negative bacteria, as well as fungal species, supports their potential application in combating infections, particularly those caused by antibiotic-resistant strains. These promising findings suggest that olive seed extracts could serve as natural antimicrobial agents, warranting further investigation into their mechanism of action, optimal formulations, and potential use in clinical or therapeutic applications.

4.3. Health benefits

This section explores the effects of olive stone by-products, particularly olive seeds, on various organs, including the heart, liver, kidneys, skin, and nervous system, as well as their impact on conditions such as diabetes and obesity (Tables 4 and 5).

4.3.1. Cardioprotective effects

Olive stones, enriched with bioactive compounds, offer substantial cardioprotective benefits, particularly through their influence on vascular health. Genedy et al. (Genedy et al., 2020) demonstrated that dietary supplementation with olive seed powder significantly improves cardiovascular health by modulating vasoactive factors. In rats on a HFFD, supplementation with 10 % olive seed nanoparticles yielded marked increases in vasodilators such as prostacyclin (113.56 ± 1.56 mmol/L *vs.* 69.79 ± 7.04 mmol/L), NO (3.99 ± 0.63 μmol/L *vs.* 15.59 ± 2.62 μmol/L), and eNOS (74.26 ± 1.28 ng/mL *vs.* 21.80 ± 2.49 ng/mL). In addition, a 5 % olive seed nanoparticle supplementation significantly decreased levels of ET-1 (34.21 ± 2.25 ng/mL *vs.* 95.70 ± 10.40 ng/mL), a potent vasoconstrictor. Maintaining a balance between vasodilators like NO and vasoconstrictors like ET-1 is essential for normal blood flow and pressure regulation; imbalances in these factors can lead to hypertension or hypotension (Meng, 2021; Sandoo et al., 2010), posing risks of heart failure, ischemia, and organ hypoperfusion revealed a reduction in pro-inflammatory markers such as VCAM-1 and E-selectin in HFFD rats treated with olive seed powder, particularly in the 10 % nanoparticle group (Genedy et al., 2020). Indeed, elevated VCAM-1 and E-selectin levels are closely linked to cardiovascular disease by promoting endothelial dysfunction and inflammation, which contribute to the progression of atherosclerosis, ischemia, and other vascular conditions (Koh and Park, 2018; Turhan et al., 2006). By reducing the expression of these adhesion molecules, olive seed supplementation potentially mitigates the risk of endothelial inflammation and associated cardiovascular complications.

In addition, the olive seed has extrathrombotic effects, supporting its use in cardiovascular health. Batçıoğlu et al. (2023) examined the

Table 4
Protective effects of olive stones on the cardiovascular and nervous systems and the liver, kidneys, and skin.

Extracts	Doses	Methods in vivo/in vitro	Results	References
			Cardiovascular effects	
Olive seed hydrolysate	2.5–10 mg/mL for 2 h	Enzymatic hydrolysis ACE-inhibitory Assays	• Peptides ≤ 3 kDa (5 mg/mL) exhibit significant antihypertensive effects, with Thermolysin yielding the highest ACE-inhibition capacity (IC_{50}=29 ± 9 μg/mL) • LLPSY, ALMSPH, and LYSPH peptides showed the most antihypertensive effect, with LLPSY as the most active, achieving 95.2 % ACE inhibition (IC_{50}=23.6 ± 0.3 μg/mL)	(Esteve et al., 2015; Vásquez-Villanueva et al., 2018)
Ethanol seed extract	50–1000 μg/mL for 5 min (37°C)	Chronolog system and turbidimetry Technique	• Seed extract showed 7.36 % better antithrombotic activity than the fruit (5.01 %) but lower than the leaf extract (23.22 %) (1 mg/mL)	(Batçıoğlu et al., 2023)
Olive seed powder	10 % for 14 days	Dexamethasone-induced diabetes rats	• Seed powder reduces the risk of atherosclerotic and coronary heart disease by lowering AI by 51 % and CRI by 40 %	(Samba Garba and Bouderbala, 2024).
Olive seed hydrolysate	400 mg/kg/day for 8 weeks	HCD rats	• Seed hydrolysate diminished AI by 32 % and CRI by 11 %	(Prados et al., 2020a)
Olive seed powder (native or nanoparticle forms)	5–10 % for 8 weeks	HFFD rats	• Nanoparticle form (10 %) significantly increased prostacyclin (1.6-fold), NO (2.1-fold), and eNOS (3.4-fold) • Nanoparticle form (10 %) significantly reduced VCAM−1 by 54 % and E-selectin by 71 % • Nanoparticle form (5 %) significantly reduced ET−1 by 64 %	(Genedy et al., 2020).
			Neuroprotective Properties	
Olive seed extract	100 mg/mL for 72 h	Zebrafish larvae	• Seed extract increased AChE activity during neurodevelopment to 162.2 % • Seed extract increased AChE activity by 40.1 % compared to the condition with neurotoxin alone (kainic acid)	(Cortés Castell et al., 2014b)
Metanolic seed extract	-	Inhibitory effects on AChE, BChE, and TYR enzymes	• Seed extract (Galega cultivar) showed the strongest inhibitory effects on AChE with IC_{50}= 30.68 ± 12.20 μg/mL (compared to 25.03 ± 3.01 μg/mL of galantamine), BChE with IC_{50}= 51.54 ± 11.81 μg/m (compared to 21.64 ± 1.22 μg/mL of galantamine), and TYR with IC_{50}= 27.42 ± 8.23 μg/mL (compared to 23.45 ± 1.06 μg/mL of kojic acid)	(Gouvinhas et al., 2022).
Olive stone extract	10–50 mg/L for 2 h	Human neuroblastoma cell line (SH-SY5Y)	• Stone extract (10 mg/L) maintains ~98 % cell viability from H_2O_2-induced cell apoptosis	(Cortés-Castell et al., 2016)
			Skin	
Seed ethanolic and aqueous extracts	-	-	• Seed extract had lower anti-inflammatory, antioxidant, anti-allergic, antibacterial, anti-melanogenesis, and collagen production effects than leaf, pulp, olive mill waste, stem, and flower	(Kishikawa et al., 2015)
			Liver	
Seed methanol extract	100 mg/kg for 90 days	HFFD rats	• Methanolic extract reduces AST by 37 %, ALT by 19 %, and LDH by 10 %	(Ben Saad et al., 2021)
Methanolic and aqueous seed extracts	250, 500, and 750 mg/kg for 14 days	Alloxan-induced diabetes in Wistar rats	• Methanolic and aqueous extracts (three doses) significantly reduce AST, ALT, and ALP levels • Aqueous extract (750 mg/kg) showed the most significant decrease in AST (37 %), ALT (78 %), and ALP (80 %) • Methanolic extract reduces the formation of fatty liver accumulation, preventing hepatic steatosis	(Akhtar et al., 2022)
Olive seed powder	10 % for 14 days	Dexamethasone-induced diabetes in rats	• Seed powder significantly improved AST by 21 %, ALT by 35 %, and ALP by 16 % • Seed powder significantly improves liver damage associated with diabetes	(Samba Garba and Bouderbala, 2024)
Olive cake	7.5 % for 28 days	HFD-induced obese rats	• Olive cake significantly improved AST (36 %), ALT (40 %), and ALP (31 %)	(Samba Garba and Bouderbala, 2022).
			Kidneys	
Methanolic and aqueous seed extracts	250, 500, and 750 mg/kg for 14 days	Alloxan-induced diabetes in Wistar rats	• Aqueous and methanol extracts improve renal function by reducing urea levels up to 10- and creatinine up to 19.7-fold • Methanolic extract (750 mg/kg) improved renal blood vessel thrombosis inflammation and alloxan-induced glomerular and tubular cell necrosis	(Akhtar et al., 2022)

ACE: angiotensin-converting enzyme; AChE: acetylcholinesterase; AI: atherogenic index; ALP: alkaline phosphatase ALT: alanine aminotransferase; AST: aspartate aminotransferase; BchE: butyrylcholinesterase; CRI: coronary risk index; eNOS: endothelial nitric oxide synthase; ET-1: endothelin-1; HCD: high cholesterol diet; HFD: high-fat diet; HFFD: high fat/high fructose diet; LDH: lactate dehydrogenase; NO: nitric oxide; TYR: tyrosinase; VCAM-1: vascular cell adhesion molecule 1.

antithrombotic activity of *O. europaea* (Halhalı olive) seed extract using a chronolog system and turbidimetry technique. Results showed that olive seed extract had a 7.36 % higher antithrombotic effect than the fruit (5.01 %), though it was less effective than the leaf extract (23.22 %), which indicates a promising role for seeds in reducing thrombus formation and enhancing circulatory health.

Additionally, olive seed hydrolysates have shown potent antihypertensive effects, largely through ACE inhibition. Vásquez-Villanueva et al. (2018) isolated peptides with significant ACE inhibitory activity, particularly the LLPSY peptide, which achieved a 95.2 % inhibition rate with an IC_{50} of 23.6 ± 0.3 μg/mL. However, gastrointestinal enzyme hydrolysis slightly reduced this inhibition by 11.6 %, suggesting partial degradation of bioactive compounds during digestion.

Moreover, Esteve et al. (2015) further investigated smaller peptides (<3 kDa), finding that Thermolysin hydrolysate had the highest antihypertensive capacity with an IC_{50} of 0.029 ± 0.009 mg/mL. The bioactive effects of these peptides persisted after *in vitro* gastrointestinal digestion, with only a minor increase in IC_{50} (0.036 mg/mL) after digestion, thus supporting their potential for use in functional foods.

Among the identified antihypertensive peptides, LLPSY, ALMSPH,

Table 5
Protective effects of olive stones on metabolic disorders.

Extracts	Doses	Methods in vivo/in vitro	Results	References
Diabetes				
Seed methanolic extract	100 mg/kg for 90 days	HFFD rats	• Methanolic extract reduces α-amylase activity by 26 % and glycemia by 27 %	(Ben Saad et al., 2021)
Methanolic and aqueous seed extracts	250, 500, and 750 mg/kg for 14 days	Alloxan-induced diabetes in Wistar rats	• Aqueous and methanolic extracts decreased α-amylase activity (1.6 mg/mL) by 82.1 % and 64.90 %, respectively • Aqueous and methanolic extracts significantly reduced glycemia • Methanolic extract dose-dependent administration increases the number of Langerhans islets in the pancreas	(Akhtar et al., 2022)
Olive seed powder (native or nanoparticle forms)	5–10 % for 8 weeks	HFFD rats	• Nanoparticle form (5 %) reduces glycemia by 40 % • Native form (5–10 %) significantly decreases insulin levels and improves insulin sensitivity, with native form (10 %) showing the most decrease by 33 % • Native or nanoparticle forms (5–10 %) decreased HOMA-IR significantly, with the native form (10 %) showing the most decrease, at 57 %	(Genedy et al., 2020)
Olive seed powder	10 % for 14 days	Dexamethasone-induced diabetes in rats	• Olive powder reduces glycemia by 20 % and HbA1c by 36 %, improving glucose tolerance	(Samba Garba and Bouderbala, 2024)
Olive seed hydrolysates	1.5 and 5 mg/mL of alkalase and papain hydrolysates for 1 h	Human epithelial cell line (Caco−2) Mouse intestinal neuroendocrine cell line (STC−1)	• Seed alkalase and papain hydrolysates (5 mg/mL) reduced DPP-IV activity by 42.9 % and 38.7 %, respectively • Seed hydrolysates increased GLP−1 level by approximately 2-fold in STC−1 culture	(Bartolomei et al., 2022)
Obesity and lipid profile				
Olive seed polyphenol extract	10–50 mg/L for 5–7 days	Mice fibroblast cell line (3T3L1)	• Seed extract (50 mg/l) decreases adipocyte differentiation by 50 %, resulting from a 5- and 1.5-fold reduction in LEP and PPARγ gene expression, respectively	(Veciana-Galindo et al., 2015)
Methanol olive stone extract	100 mg/kg for 90 days	HFFD-induced obese/diabetes rats	• Stone extract decreases lipase activity (37 %), TC (17 %), and LDL-C (48 %) • Stone extract increases HDL-C by 87 % • Stone extract reduces body weight by 22 %	(Ben Saad et al., 2021)
Olive seed powder	10 % for 14 days	Dexamethasone-induced diabetes in rats	• Seed powder decreased TC (29 %), LDL-C (57 %), and TG (19 %) levels while increasing HDL-C levels by 23 %	(Samba Garba and Bouderbala, 2024).
Olive cake	7.5 % for 28 days	HFD-induced obese rats	• Olive cake significantly decreased lipase activity (38 %), TC, TAGs, LDL-C, and increased HDL-C	(Samba Garba and Bouderbala, 2022)
Olive seed powder (native or nanoparticle forms)	5–10 % for 8 weeks	HFFD rats	• Nanoparticle form (10 %) decreases intestinal lipase activity (56 %), TC (25 %), TAGs (56 %), LDL-C (64 %), and increases HDL-C by 23 %	(Genedy et al., 2020)
Seed hydrolysate	400 mg/kg/day for 8 weeks	HCD rats *In vitro* assay of hypocholesterolemic activity	• Seed hydrolysate lowered TC by 25 % and LDL-C by 43 % • Peptides from 20 olive varieties inhibited HMG-CoA reductase by up to 40 % and reduced micellar cholesterol solubility by 11–49 % • Certain peptides decreased bile acid binding and cholesterol esterase activity	(Prados et al., 2020a, 2020b)

DPP-IV: dipeptidyl peptidase-IV; GLP-1: glucagon-like peptide-1; HbA1c: glycated hemoglobin A1c; HCD: high cholesterol diet; HFD: high-fat diet; HFFD: high-fat/high-fructose diet; HDL-C: high-density lipoprotein cholesterol; HMG-CoA: 3-hydroxy-3-methylglutaryl-coenzyme A; LDL-C: low-density lipoprotein cholesterol; LEP: leptin; PPARγ: peroxisome proliferator-activated receptor γ; TAG: triacylglycerol; TC: total cholesterol.

and LYSPH showed significant effects due to the presence of proline in the last three amino acids. Proline is known to contribute to ACE inhibition, as it enhances the interaction between peptides and ACE, blocking the enzyme's activity. These findings highlight the potential of olive seed hydrolysates as a source of antihypertensive peptides, offering an alternative strategy for managing hypertension through dietary means.

4.3.2. Hepatoprotective effects

The hepatoprotective properties of olive seed extracts have been substantiated in studies that observed improvements in liver enzyme profiles. Ben Saad et al. (2021) demonstrated that methanol extract of Olea meski seeds (100 mg/kg over 90 days) decreased AST, ALT, and LDH levels in rats on HFFD, indicative of improved liver function. Likewise, Akhtar et al. (2022) reported that methanol and aqueous extracts of *O. europaea* subsp. *Cuspidata* (Indian olive) significantly reduced AST, ALT, and ALP levels in Wistar rats when administered in doses ranging from 250 to 750 mg/kg. Additionally, a short-term intervention (14 days) with 10 % olive seed powder in dexamethasone-treated rats significantly improved liver enzyme profiles, further supporting the protective effects of olive seed components (Samba Garba and Bouderbala, 2024).

Apart from enzymatic improvements, olive seed extracts were found to mitigate hepatic steatosis, preventing excessive fatty liver accumulation and indicating protection against NAFLD (Akhtar et al., 2022; Ben Saad et al., 2021). Moreover, Samba Garba and Bouderbala, (2022) found that olive cake consumption ameliorated liver function markers in obese rats on an HFD, showing reductions in AST (36 %), ALT (40 %), and ALP (31 %) levels compared to untreated controls. These studies suggest that olive seed and its derivatives may serve as valuable adjuncts in preventing or managing liver disease by reducing fat accumulation and improving liver enzyme function.

4.3.3. Renoprotective effects

Olive seed extracts also exhibit renoprotective effects, potentially enhancing kidney function. Ben Saad et al. (2021) found that methanol extracts from *Olea meski* seeds (250–750 mg/kg) improved renal function by reducing urea and creatinine levels in rats, suggesting enhanced filtration and excretory performance. Akhtar et al. (2022) observed similar benefits with methanol and aqueous extracts of *O. europaea* subsp. *Cuspidata*, where doses between 250 and 750 mg/kg lowered urea and creatinine in Wistar rats, indicative of improved renal health.

Furthermore, high doses of methanol seed extract (750 mg/kg) demonstrated protective effects against renal inflammation and necrosis

in glomerular and tubular cells induced by alloxan, which is known to cause diabetic-like kidney damage in rats (Akhtar et al., 2022). These findings highlight olive seed's potential in ameliorating nephrotoxic damage, emphasizing its relevance in preventing kidney-related complications, especially in diabetes and obesity.

4.3.4. Neuroprotective properties

Olive seed extracts exhibit significant neuroprotective and anti-neurodegenerative effects. *In vitro* studies indicate that olive stone extracts at 10 mg/L concentrations prevent apoptosis in SH-SY5Y neuroblastoma cells exposed to H_2O_2, maintaining ~98 % cell viability by modulating the Bax/Bcl-2 ratio in favor of cell survival (Cortés-Castell et al., 2016). This antioxidant property of olive stone phenolics supports their role in counteracting oxidative stress, a major factor in neurodegenerative diseases.

Other studies show that polyphenols from olive pits possess strong inhibitory activity against key enzymes involved in neurodegeneration, such as AChE, BChE, and tyrosinase. Indeed, Galega cultivar extracts exhibited high inhibition rates for AChE ($IC_{50} = 30.68 \pm 12.20$ μg/mL), BChE ($IC_{50} = 51.54 \pm 11.81$ μg/mL), and tyrosinase ($IC_{50} = 27.42 \pm 8.23$ μg/mL), with phenolic compounds like nüzhenide, tyrosol, rutin, luteolin-7-glucoside, oleuropein, and ligstroside being potential contributors (Gouvinhas et al., 2022). Indeed, Sung et al. (2006) reported that nüzhenide has significative neuroprotective effects and can protect SH-SY5Y human neuroblastoma cells against 6-hydroxydopamine-induced neurotoxicity. Cobrançosa and Picual cultivars also showed enzyme inhibition but were less potent than Galega ($p < 0.05$) (Gouvinhas et al., 2022).

In neurodevelopmental models, olive seed extracts enhanced AChE activity by 162.2 % in zebrafish larvae, suggesting neurodevelopmental benefits. Furthermore, in neurotoxicity assays, larvae co-treated with olive seed extract and kainic acid demonstrated 140.1 % AChE activity, outperforming those treated with kainic acid alone, thereby confirming the protective action against excitotoxic damage (Cortés Castell et al., 2014b). These studies support olive seed extracts as promising sources of neuroprotective agents for neurodegenerative and neurodevelopmental disorders, including Alzheimer's and Parkinson's (Cortés Castell et al., 2014b; Gouvinhas et al., 2022).

4.3.5. Dermatological applications

Olive seed extracts have potential applications in dermatology, although research primarily focuses on olive leaf, fruit pulp, and other by-products. Kishikawa et al. (2015) demonstrated that ethanolic leaf extracts inhibit melanin production in B16 melanoma cells, showing promise for skin-whitening applications. Additionally, both leaf and fruit pulp extracts stimulated collagen production and fibroblast proliferation, contributing to skin elasticity and anti-aging effects without cytotoxicity.

Ethanolic extracts from olive mill waste exhibited anti-allergic effects by reducing granule release in basophilic leukemia cells (RBL-2H3), suggesting potential applications for sensitive skin (Kishikawa et al., 2015). Moreover, olive seed extracts were noted to have lower total phenolic content than other parts of the olive tree, such as stems and leaves, possibly affecting their bioactivity (Kishikawa et al., 2015). However, polyphenols like oleuropein and hydroxytyrosol, present in various olive by-products, are indispensable for skin health due to their anti-aging, moisturizing, anti-inflammatory, and UV-protective properties (Michalak, 2022).

Oleuropein specifically has shown antioxidant and anti-melanogenesis properties, potentially aiding in hyperpigmentation treatment (Kishikawa et al., 2015). It also inhibits MMP-13 expression, protecting the skin from UV-induced collagen degradation and dermal thinning (Sumiyoshi and Kimura, 2010). Similarly, hydroxytyrosol has anti-inflammatory and radioprotective actions, reducing keratinocyte DNA damage and death due to oxidative stress. It modulates heme oxygenase-1 expression, enhancing skin cell resilience against oxidative injuries (Rafehi et al., 2012). These findings underscore the potential of olive seed and other olive by-products as sustainable sources of bioactive compounds for dermatological use, from anti-aging creams to sun-protective formulations.

4.3.6. Protection from other diseases

Olive stone by-products, especially olive seeds, have shown potential in managing various metabolic diseases, including diabetes, obesity, and hyperlipidemia, and in promoting intestinal health through modulation of the gut microbiota (Table 5).

4.3.6.1. Type 2 Diabetes. Studies have highlighted the potential of olive seeds in type 2 diabetes management. Ben Saad et al. (Ben Saad et al., 2021) reported that methanolic extracts from *O. meski* seeds inhibit α-amylase activity by 26 % (IC_{50}=1.04 mg/mL), thereby slowing carbohydrate breakdown. In HFFD-rats, a 90-day regimen of 100 mg/kg methanolic olive stone extract led to a 27 % reduction in blood glucose levels compared to untreated controls. Genedy et al. (Genedy et al., 2020) observed that 5–10 % olive seed powder, both in native and nanoparticle forms, significantly reduced glucose and insulin levels and improved insulin sensitivity. In HFFD rats, supplementation with 5 % olive seed powder resulted in a considerable serum glucose reduction (94.43 ± 0.30 mg/dL *vs.* 156.46 ± 1.80 mg/dL), while insulin levels decreased more significantly in the native form (5–10 %) with a HOMA-IR index drop of 52–56 %. The hypoglycemic effect is attributed to the significant phenolic compound nüzhenide and also to hydroxytyrosol, oleuropein aglycon, oleuropein, and triterpene oleanolic acid (Ben Saad et al., 2021; Genedy et al., 2020; Montó et al., 2014). Additionally, olive stone fiber has demonstrated benefits in modulating biochemical markers such as HbA1c, fasting glucose, and insulin levels (Mao et al., 2021). A study also found that a 10 % olive seed powder supplementation for 14 days decreased blood glucose by 20 % and HbA1c by 36 % in dexamethasone-induced diabetic rats, indicating improved glucose tolerance and preservation of pancreatic islet architecture (Samba Garba and Bouderbala, 2024). Indeed, seed extracts restored Langerhans islets and protected against pancreatic damage (Akhtar et al., 2022; Samba Garba and Bouderbala, 2024). Additionally, olive seed proteins offer bioactive peptides with potential antidiabetic properties. Recent findings suggest that olive seed hydrolysates inhibit DPP-IV and increase GLP-1 levels, thereby enhancing insulin secretion. Specifically, alkalase hydrolysates decreased DPP-IV activity by 42.9 % and papain hydrolysates by 38.7 % at a concentration of 5 mg/mL, with GLP-1 levels reaching 660.7 pM and 613.4 pM, respectively (Bartolomei et al., 2022). These hydrolysates present valuable potential for nutraceutical and functional food applications targeting diabetes prevention.

4.3.6.2. Obesity and lipid profile. Olive seed extract has shown anti-adipogenic activity, as reported by Veciana-Galindo et al. (Veciana-Galindo et al., 2015), who found a 50 % reduction in adipocyte differentiation and a decrease in *leptin* and *PPARγ* gene expression in pre-adipocytes treated with 50 mg/L olive seed polyphenols. In HFFD rats, Ben Saad et al. (2021) found that methanol olive stone extract decreased intestinal lipase activity by 37 % (IC_{50}=2.13 mg/mL), which subsequently led to a 17 % reduction in TC and a 48 % reduction in LDL-C while preventing a 51 % decline in HDL-C in diabetic rats. This extract also mitigated obesity, resulting in a 22 % body weight reduction in treated rats. Indeed, nüzhenide has been shown to have anti-obesity effects on HFD mice. Liu et al. (2014) reported that a 30 mg/kg of purified extract of nüzhenide from *Ligustrum lucidum* fruits induced a significant 32 % decrease in body weight over 6 weeks compared to the controls. Moreover, fiber from olive stones is also linked to improved lipid profiles, including lower TC and LDL-C, through increased satiety and modulation of biochemical markers (Jenko Pražnikar et al., 2023; Lu et al., 2023). In a dexamethasone-induced dyslipidemia model, 10 % olive seed powder supplementation over 14 days improved lipid

markers by reducing TC, LDL-C, and TAGs while boosting HDL-C levels by 23 % (Samba Garba and Bouderbala, 2024). Additionally, Genedy et al. (2020) found that a high dose (10 %) of olive seed nanoparticles markedly improved lipid profile in HFFD rats, with reductions in TC, TAGs, LDL-C, VLDL-C levels, and lipase activity, while elevating HDL-C levels to near-healthy values (Ben Saad et al., 2021; Genedy et al., 2020). In fact, nüzhenide may have lipid-lowering effects. *In vitro* studies reported that nüzhenide extracted from *F. mandshurica* seeds showed significant pancreatic lipase inhibitory activity at a concentration of 100 μM, with an inhibition rate of 43.45 % (Guo et al., 2020). Moreover, these effects can also be attributed to other minor compounds like hydroxytyrosol, oleuropein, ferulic acid, naringenin, oleuropein aglycon, gallic acid, and oleanolic acid (Ben Saad et al., 2021; Genedy et al., 2020). Prados et al. (Prados et al., 2020a) reported that 400 mg/kg/day olive seed hydrolysate supplementation for eight weeks in HCD rats resulted in a 25 % decrease in TC and a 43 % reduction in LDL-C, partly through inhibition of cholesterol micellar solubility by 11–49 % and bile acid binding (10 %), while inhibiting the pancreatic enzyme cholesterol esterase by 15–31 % (Prados et al., 2020b). Besides, seed hydrolysate derived from the Manzanilla olive variety inhibits pancreatic lipase by 50 % (Prados et al., 2018). Olive seed hydrolysate also reduces HMG-CoA reductase activity by up to 40 % at a peptide concentration of 26.7 mg/mL, showing comparable effects to the TC-lowering agent pravastatin (Prados et al., 2020a). HMG-CoA reductase catalyzes the conversion of HMG-CoA to mevalonate in the presence of the NADPH cofactor, a key initial step in the cholesterol biosynthesis pathway (DeBose-Boyd, 2008).

4.3.6.3. Gut Microbiota. The high insoluble fiber content in olive stones contributes significantly to gastrointestinal health and offers protective benefits against various diseases, including certain cancers and obesity-related disorders. Insoluble dietary fiber, strongly linked to improved digestive system function, is essential in reducing cancer risk and supporting weight management (Reynolds et al., 2020). The recommended daily intake for adults is generally 25–30 g of dietary fiber, although exact requirements may vary by country, age, and gender (McKeown et al., 2022). Olive stone fiber improves intestinal health by enhancing transit time, increasing stool volume, and retaining water, which aids in preventing constipation (Khorasaniha et al., 2023).

Moreover, the gut microbiota ferments dietary fiber to produce SCFAs, such as butyrate, propionate, and acetate, which serve critical roles in gastrointestinal and systemic health. SCFAs support the proliferation of colonocytes, the cells lining the colon, and reinforce the integrity of the intestinal barrier (Fusco et al., 2023). Furthermore, these SCFAs exert potent anti-inflammatory effects, decreasing the production of pro-inflammatory cytokines by immune cells, including neutrophils, macrophages, and NK cells, while also promoting apoptosis in cancer cells (Son and Cho, 2023). For instance, butyrate and propionate reduce inflammation by inhibiting the activation of the NF-κB pathway, particularly by blocking the acetylation of the p65 subunit, which leads to increased production of anti-inflammatory cytokines such as IL-10, IL-18, and TGF-β (Liu et al., 2023). In parallel, these SCFAs reduce pro-inflammatory cytokines and chemokines, including TNF-α, TNF-β, IL-1β, and MCP-1 (monocyte chemoattractant protein-1), further reducing inflammation. Butyrate also exerts additional anti-inflammatory effects by inhibiting the formation and activation of the NLRP3 inflammasome, a multiprotein complex involved in inflammatory responses (Liu et al., 2023).

Dietary fiber consumption has shown promise in decreasing the risk of various cancers, including colorectal, breast, and prostate cancers (Hu et al., 2023). The chemopreventive effects of SCFAs, particularly butyrate, stem from their ability to inhibit HDAC activity, which plays a vital role in gene expression. Through this mechanism, SCFAs modulate the expression of genes involved in cancer regulation by upregulating tumor suppressor genes (such as *p53* and *p21*) and downregulating oncogenes (such as *HIF-1α*, *EGFR*, and *CSE1L*), as well as by affecting key pathways essential for tumor progression. These include the Wnt/β-catenin, PI3K/Akt/mTOR, MAPK, Ras/Raf/ERK, and Notch signaling pathways, which are critical for cell proliferation and apoptosis resistance (Feitelson et al., 2023; Son and Cho, 2023).

Moreover, the insoluble fiber in olive stones accelerates gastrointestinal transit time, reducing mucosal exposure to potentially toxic compounds, including carcinogens (Novianti et al., 2023). By decreasing transit time, the fiber limits the interaction between carcinogens and the intestinal lining, offering an additional layer of protection against the development of colorectal cancer (Novianti et al., 2023).

5. Applications in agriculture and horticulture

Agricultural by-products, often disposed of through burial or burning, can lead to environmental consequences. A promising alternative involves repurposing these by-products as substrates in agriculture and horticulture. Olive stones exhibit ideal substrate properties due to their physical stability, high porosity, and neutral pH. Hydrothermally treated olive stone substrates have been shown to enhance germination rates and the shoot-root development of *Raphanus sativus* L. Moreover, olive stone-based substrates support the growth of essential horticultural crops, including tomatoes, cucumbers, and peppers, highlighting their potential in agricultural practices (Karakaş et al., 2017). Future studies into various stone and peat mixtures for substrate granulation could further optimize these applications. Academic research has established a solid foundation for the practical use of olive stone by-products in agriculture, producing substrates with characteristics comparable to peat. This repurposing of olive stones presents promising opportunities for companies to pursue sustainability, corporate responsibility, and product diversification (Sciubba et al., 2020).

5.1. Soil amendment and fertilizer

In Mediterranean regions, the olive oil industry generates an abundance of by-products (Bourgeon et al., 2018). Though often used as animal feed, olive stones' high lignin content, and favorable nutrient composition, including low salinity, high organic matter, and an optimal C/N ratio, make them ideal candidates for soil amendments and slow-release fertilizers (Gul et al., 2021; Rodríguez-Gutiérrez et al., 2014). Studies suggest that incorporating olive stones into the soil may enhance soil quality and fertility while simultaneously sequestering carbon, potentially reducing pollution in the Mediterranean's agriculture-intensive zones (Ayman and Fawzy, 2023; Fornes et al., 2024). However, most research has focused on pot experiments or sandy soils; thus, broader studies examining crop yield effects and nutrient leaching are still necessary (Galic and Bogunovic, 2018).

Olive stones processed for soil amendments are shredded into granules, with the majority having diameters between 2 and 5 mm (73.9 % by weight) (Cuevas et al., 2019). This size prevents clogging in irrigation systems while supporting percolation and micropore formation, enhancing soil structure. The material's low specific weight compared to traditional granular fertilizers aids in moisture retention. The shredding process involves separating dust and shavings by air suction to achieve a pure substrate (Castells et al., 2023). Unlike conventional soil-based fertilizers, olive stone granules create stable percolation channels and soil aggregates, contributing to long-lasting soil improvement (Valvez et al., 2021).

5.2. Plant growth promotion

Rooting stem cuttings are essential for clonal propagation in horticulture, yet many species face challenges in forming adventitious roots. Environmental and soil-related factors, such as moisture and nutrient availability, can significantly impact rooting success (Marschner and Rengel, 2023). Olive stones, whether used alone or mixed with peat,

have shown promise as a supportive growth medium for various plant species (Chrysargyris et al., 2019; Karakaş et al., 2017), as they offer high lignin and carbon content that can improve substrate stability and nutrient release (Sciubba et al., 2020). Traditionally, the disposal of olive stones through landfilling or burning poses environmental risks, but sustainable practices like co-composting, thermochemical treatments, or use as plant substrates offer eco-friendly alternatives (Valvez et al., 2021).

However, few studies have specifically investigated olive stone-based propagation systems for rooting cuttings *in vitro* (Montilon et al., 2022). Additional research on ground olive stone as a growing substrate could reveal new applications for plant propagation, particularly in ornamental and horticultural industries. By exploring the rooting efficacy of various plant species within olive stone substrates, this approach may uncover innovative solutions to enhance nursery production and agricultural sustainability.

In summary, olive stones provide an ecological and economic value in agriculture and horticulture, offering a sustainable alternative to polluting waste management practices and diversifying applications within a sustainability framework.

6. Industrial applications of olive stone by-product

The by-products of olive oil production, collectively known as olive pomace, include the skin, pulp, and stone of the olive fruit (Sánchez Moral and Ruiz Méndez, 2006). Traditionally, this pomace has been either discarded as waste or minimally repurposed, yet it represents a valuable resource with significant industrial potential. The olive stone, in particular, can be transformed into olive stone flour, which has found innovative uses in the food industry (Naghmouchi et al., 2015). For instance, studies have demonstrated that olive stone flour can enhance the nutritional value of bread by adding dietary fiber and improving the texture, making it crunchier and more crumbly (Simsek and Süfer, 2022). Olive stone flour may also undergo microbial fermentation to produce extracts that act as bio-ingredients for bread-making, substituting up to 75 % of traditional flour and enriching bakery products with antioxidants (Bolek, 2020; Cuevas et al., 2009).

Beyond food, olive stones can be converted into renewable energy in the form of solid pellets (Matos et al., 2010), characterized by their low moisture content (6.5 %), high density (774.2 kg/m^3), and calorific value (4386 kcal/kg). When burned in controlled experimental reactors, these pellets have demonstrated their feasibility as a bioenergy source, offering an efficient alternative to fossil fuels while reducing environmental impact (Cui et al., 2021). Additionally, the physical properties of olive stones, including their lignocellulosic structure (Rodríguez-Gutiérrez et al., 2014), make them suitable for use in wood composite boards bonded with high-performance adhesives such as isocyanate or phenol-formaldehyde. These boards meet European standards (EN) for particleboards and strand boards, showing superior strength and durability (Valvez et al., 2021).

6.1. Energy production

The global demand for renewable energy sources has spurred interest in biomass fuels, especially in regions with high agricultural production, like the Mediterranean, where olive oil industries produce large quantities of by-products annually. Olive pomace and olive stones, by-products of the olive oil extraction process, have gained particular attention due to their high lignocellulosic content and availability (Rodríguez-Gutiérrez et al., 2014). Large quantities of olive stones are produced globally each year (García Martín et al., 2020), highlighting the importance of converting this biomass into energy. This approach offers a sustainable solution for waste management and renewable energy generation. This conversion aligns with environmental policies focused on transforming waste into a valuable resource, addressing disposal issues, and supporting energy needs sustainably.

Bio-conversion of olive stones into biofuels such as solid biomass, bioethanol, or biogas offers several advantages (Mata-Sánchez et al., 2014). Biomass fuels derived from olive stones are low-cost, sustainable, and widely accessible in regions where olive oil production is prevalent. Energy generated from biomass not only reduces dependency on fossil fuels but also cuts greenhouse gas emissions, thereby benefiting the environment (Bridgwater, 2006). Olive stone biomass, specifically, is emerging as a prime candidate for replacing fossil fuels within a circular bioeconomy framework, addressing both waste disposal and renewable energy production. The high calorific value of olive stone biomass makes it suitable for combustion, converting it into heat and power through bio-conversion processes that hold significant potential for countries facing energy shortages and waste management challenges.

6.2. Construction materials

The growing need for eco-friendly construction materials has driven research into the use of agricultural by-products as sustainable replacements for conventional construction aggregates (Garg and Jain, 2014). Olive stones, which represent a substantial portion of olive pomace, are being investigated for their potential use in green building materials, such as bio-based bricks and eco-friendly concrete composites (San Vicente-Navarro et al., 2023). Their high lignocellulosic content contributes to strength and durability (Rodríguez-Gutiérrez et al., 2014), allowing them to perform well as a natural aggregate in various construction applications.

In Mediterranean countries, where the two-phase olive oil extraction system is increasingly replacing the traditional three-phase process, a significant amount of wet olive pomace is generated (Sibaja et al., 2005). This by-product poses an environmental challenge due to its disposal requirements (Los Santos-Ortega et al., 2025). However, reusing olive pomace in construction materials offers a sustainable solution that minimizes environmental impact while addressing construction demand. When incorporated into bio-composites (Valvez et al., 2021), olive stones serve as an eco-friendly alternative to synthetic materials, supporting the circular economy within the construction industry by promoting resource efficiency and reducing waste (Bai et al., 2023; Binici and Aksogan, 2016).

In summary, olive stone by-products are gaining attention for their multifaceted industrial applications. As a source of biofuel, they offer a sustainable energy alternative that aligns with global goals to reduce fossil fuel dependence. Additionally, olive stones present valuable opportunities in food and construction industries, enhancing both nutritional value in bakery products and providing eco-friendly materials for infrastructure. Through the valorization of olive stone waste, the olive oil industry is contributing to a circular economy where agricultural residues are repurposed for diverse and sustainable industrial uses.

7. Challenges and future directions of olive stone by-product utilization

As with many forms of agricultural biowaste, innovative methods for the use and disposal of olive stone by-products should be prioritized to maximize sustainability. Although extensive research has examined various applications of olive stone by-products, including the extraction and utilization of bioactive compounds, additional strategies to enhance its value are still needed. For instance, olive stone by-products can be used as a soil improver (Gibbert et al., 2024; Payá et al., 2020), a substrate for mushroom cultivation, or even for vermicomposting. These applications align well with the objectives of sustainable agriculture, especially in regions reliant on olive cultivation, where maximizing olive stone by-product utilization can support the local economy and improve the overall competitiveness of agriculture in these areas.

To achieve comprehensive valorization of olive stone by-products throughout the entire value chain, global projects that integrate industrial partnerships and multidisciplinary approaches are essential. Such

projects should consider not only technical feasibility but also economic, social, and regulatory factors that may impact implementation on both local and national levels. Indeed, developing applications based on olive stone by-products for mushroom cultivation or bioactive compound extraction in collaboration with social or rural cooperatives could help expand economic opportunities in these regions. This approach may also be more socially acceptable, as it does not compete with food or livestock resources, reducing potential resistance.

The concept of a circular economy, reinforced by biorefinery frameworks, is critical for minimizing carbon footprints and promoting a "zero-waste" approach. By transforming low-value or traditionally discarded by-products like olive stone by-products, waste oil, and olive mill wastewater into high-value products, the agricultural sector can transition to waste-free bioprocesses (Galiatsatou et al., 2002; Mohammadnejad et al., 2021). Such an approach not only enhances environmental sustainability by promoting a more rational use of resources but also aligns with broader sustainability goals (Valvez et al., 2021).

8. Technological limitations

Technological limitations present a significant challenge to the efficient utilization of olive stone by-products (Restuccia et al., 2018). These limitations are often tied to the inherent properties of the by-product, including hazardous characteristics, the absence of suitable technological processes, or the limited adoption of existing technologies for on-site by-product management. Addressing these challenges requires a comprehensive understanding of each limitation to devise targeted solutions and strategies for overcoming them (Frangipane et al., 2022).

Some of the main issues associated with olive stone by-product utilization are related to dust generation and bioaerosol formation during grinding processes (Viteri et al., 2024), which pose health risks to workers and require stringent health hazard analyses. In response, current treatment technologies emphasize processes such as cooling and wet grinding to mitigate dust production (Plasquy et al., 2021). Additionally, dust suppression methods have been explored, including the incorporation of additives (Sakar and Gharby, 2022), such as sterols, salts, and biogenic agents within the olive oil industry, to improve handling and safety in later processing stages.

For new olive stone by-products, certain technological properties may complicate their application across diverse markets, particularly due to variations in regulatory standards. In the European Union (EU), for example, the composting of olive stones lacks standardized legislative frameworks or specific regulatory guidance (Conte et al., 2020). Without clear legislation, the ability to capitalize on the potential of olive stone by-products, particularly for composting and soil amendment purposes, remains limited, which further hinders their adoption in agriculture and industry.

9. Regulatory constraints

Regulatory constraints significantly influence the sustainable integration of olive stone by-product across industries, particularly in sectors like construction and energy. Within the EU, strict regulations on the use of new raw materials pose barriers to the incorporation of agricultural by-products as fillers or admixtures. Despite research showing that olive stone can serve as a viable additive in plaster matrices (Barreca and Fichera, 2013; Liuzzi et al., 2020), its application remains constrained by regulatory conventions that lack precedents for agricultural by-product utilization. Therefore, to facilitate broader adoption, it is necessary to establish a harmonized European standard that clearly defines olive stone utilization in construction materials.

In the energy sector, the regulatory framework governing olive stone by-products as a biomass source varies across EU member states, affecting its widespread use (Conte et al., 2020). Spain, for instance, has regulations aimed at reducing abandonment practices of agricultural by-products (Carmona-Torres et al., 2023), yet these same regulations impose restrictions on burning olive stone with traditional fuels in older combustion plants. The integration of olive stones into newer plants is encouraged, but achieving significant implementation would require policy incentives to support either retrofitting older plants or building new ones optimized for clean olive stone combustion. Similarly, other EU countries face regulatory challenges in transitioning olive stone usage for energy while adhering to the traditional energy infrastructure (Mata-Sánchez et al., 2014). However, efforts across the EU increasingly focus on clean and sustainable conversion pathways for agro-industrial by-products, promoting a circular bioeconomy from agricultural residues through treatment processes to potential bio-products (Lo Giudice et al., 2021).

In summary, while olive stone by-product presents promising avenues for sustainable and economic benefits across multiple industries, a concerted effort to overcome technological and regulatory barriers is essential. Addressing these challenges through targeted research, regulatory reform, and collaborative projects would foster the full potential of olive stone by-products, advancing both environmental sustainability and economic development in olive-producing regions.

10. Conclusions and future perspectives

The sustainable utilization of olive stone by-products offers a unique intersection of health, energy, and environmental benefits, highlighting its multidimensional potential in various sectors. The bioactive compounds present in olive stones, notably those with antioxidant and anti-inflammatory properties, demonstrate significant promise for promoting health by potentially reducing the risks of cardiovascular and neurological diseases. These findings reinforce the role of olive stone by-products as a valuable source of natural compounds that could be explored further in functional food and nutraceutical industries, contributing to public health advancements. Moreover, the conversion of olive stone by-products into renewable biomass energy carriers addresses both economic and environmental imperatives. With its high calorific value and low moisture content, olive stone waste is well-suited for energy applications, providing a sustainable and efficient fuel option that supports the circular economy model. By repurposing this agricultural waste as a biofuel, the olive industry can reduce its carbon footprint and dependence on fossil fuels, thus contributing to broader climate goals.

Transforming olive stone waste into valuable resources can greatly ease waste management challenges in the olive oil industry, which generates large amounts of waste annually. This approach minimizes environmental pollution and reduces strain on natural resources, providing eco-friendly alternatives across agriculture, construction, and energy sectors. Such a zero-waste strategy aligns with sustainable development goals by enhancing resource efficiency and environmental protection. Beyond environmental and health benefits, the economic potential of olive stone by-products is substantial. Converting this waste into marketable products could boost profitability for olive producers and strengthen the economy of olive-growing regions. Partnering with local cooperatives or social enterprises in these applications could create jobs and promote regional economic growth.

Moving forward, further research and innovation are crucial to realizing the full potential of olive stone by-products. Overcoming technological and regulatory barriers will be essential for scaling these applications, supported by collaborative efforts across industries, academia, and policymakers.

In conclusion, the valorization of olive stone by-products supports both environmental and economic sustainability, facilitating the olive oil industry's shift toward a greener, resilient future. Investing in this promising field will help achieve sustainable development goals, making olive stone by-products a cornerstone of sustainable practices.

CRediT authorship contribution statement

Meriem El Fessikh: Methodology, Data curation, Analysis, Investigation, Writing-Original draft preparation. **Hamza Elhrech:** Methodology, Investigation, Data Curation. **Amina El Yahyaoui El Idrissi:** Methodology, Investigation, Data Curation. **Learn-Han Lee:** Supervision, Methodology, Investigation, review & editing. **Waleed Al Abdulmonem:** Methodology, Supervision, Funding, and Review & editing. **Nasreddine El Omari:** Investigation and Review & editing. **Abdelhakim Bouyahya:** Conceptualization, Methodology, Software, Validation, Supervision, and Review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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